



Introduction to Python Programming

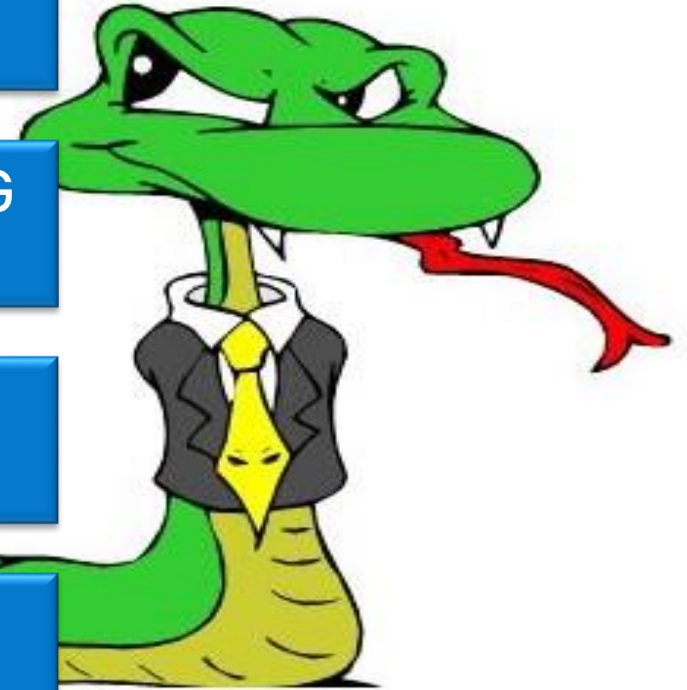
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AKSHAY GAIKWAD
PROFESSOR
V2V CLASSES

Python

Capsule

- HISTORY OF PYTHON
- FEATURES OF PYTHON
- INSTALLING PYTHON AND RUNNING A PROGRAM
- DEBUGGING THE PROGRAM
- DIFFERENCE BETWEEN BRACES, BRACKETS AND PARENTHESES



1.

HISTORY OF PYTHON PROGRAMMING

History of Python Programming

- ▷ Python is a high level programming language.
- ▷ Python was developed by Guido van Rossum.
- ▷ Python language was first released in 1991
- ▷ Python was named after Monty Python's Flying Circus BBC show
- ▷ Now a day's python is becoming popular and it is having a strong community
- ▷ In 1994, Python 1.0 was released with new features
- ▷ And then python 2.0 and after that python 3.0 was released to rectify the flaw in the language



2.

FEATURES OF PYTHON PROGRAMMING

Features of Python programming

High Level Programming Language

- ✓ High language means , this is separate from the low processor operations
- ✓ High level language is closer to Human Language than machine language

Python is a free and open source language

- ✓ Python is free, there is no need to pay money for installing python and one can easily download it from the website
- ✓ Python is an open source language, it makes the python popular

Python is simple

- ✓ In python there is no need of declaring the variables and in a single statement the users can express the complex operations
- ✓ It uses indentation for statement grouping instead of curly braces
- ✓ Python offers more structure and support for lengthy or big programs than batch files or shell script

Error checking and high level data type

- ✓ Python offers more effective an error checking than C. Python is having built in level data types for example dictionaries and flexible arrays.
- ✓ If we compare python with AWK and Perl, Python has more general data types and because of this python is applicable to bigger problem domain.

Features of Python programming

Platform Independent Language

- ✓ Python can run on any operating systems like Windows, Linux, Unix , DOS, etc . Python is interpreted Language , it saves time as it does not need compilation and linking.
- ✓ The interpreter can be utilized intelligently, which makes it easy to try different things with elements of the language , to write disposed of programs , or to test function in the midst of bottom up program development. It resembles a convenient work area calculator for clients

Python is User Friendly

- ✓ Python is having the features of abstraction and hiding of the details which makes it more user friendly. In python we can divide the programs into modules that can be used in other programs.
- ✓ .python provides interface to the GUI kit such as TK

Python is Extensible

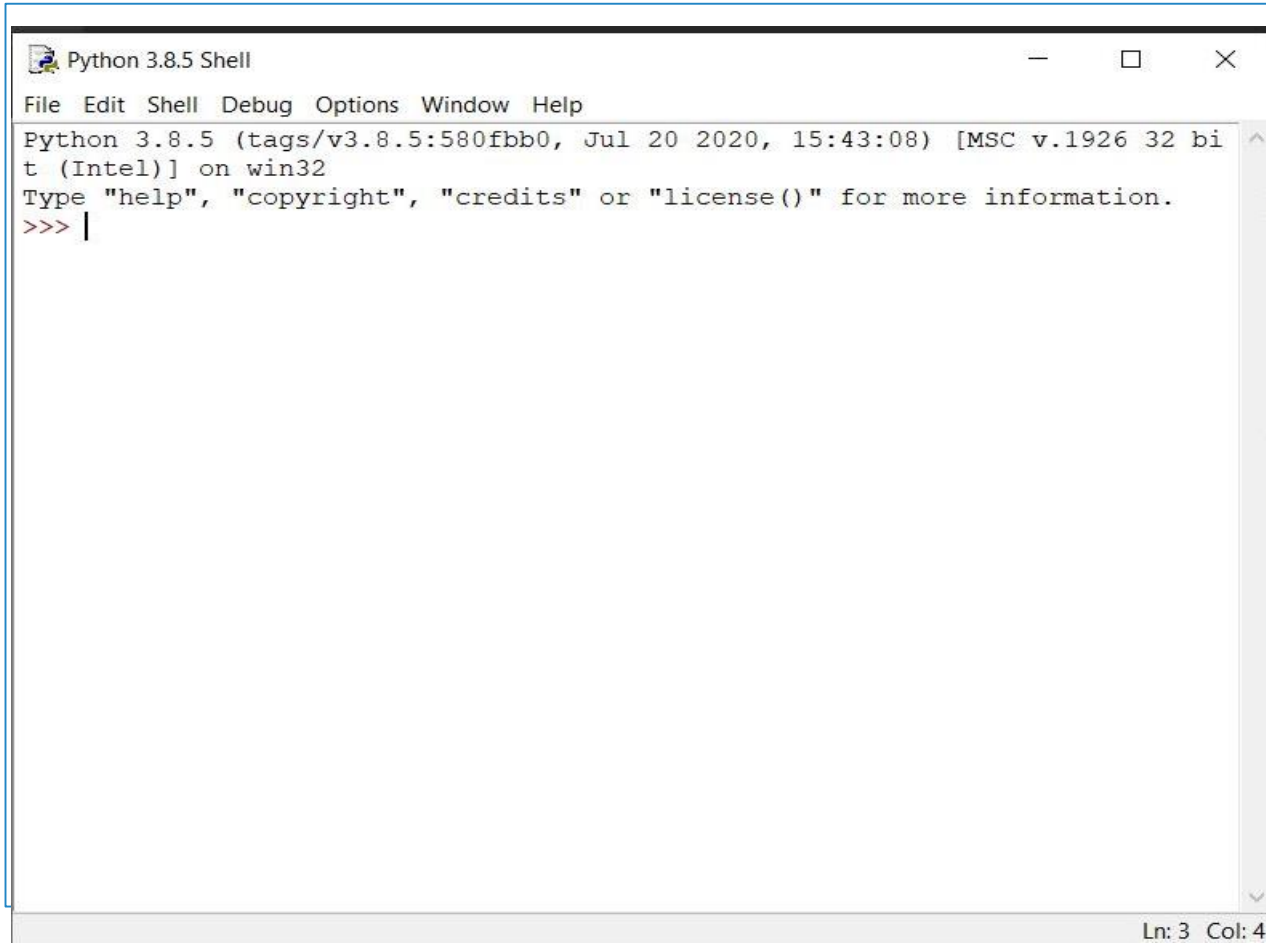
- ✓ If you have good knowledge of C programming then it is easy to add new built in function or method or module to the interpreter.
- ✓ If you are writing application in C and you are linking the python interpreter into an application then you can use it is an extension or command language for that application.

Python is object oriented

- ✓ Python supports the object oriented programming concept. But OOP is optional in python
- ✓ But whenever the user feels like using the OOPS concept user can use

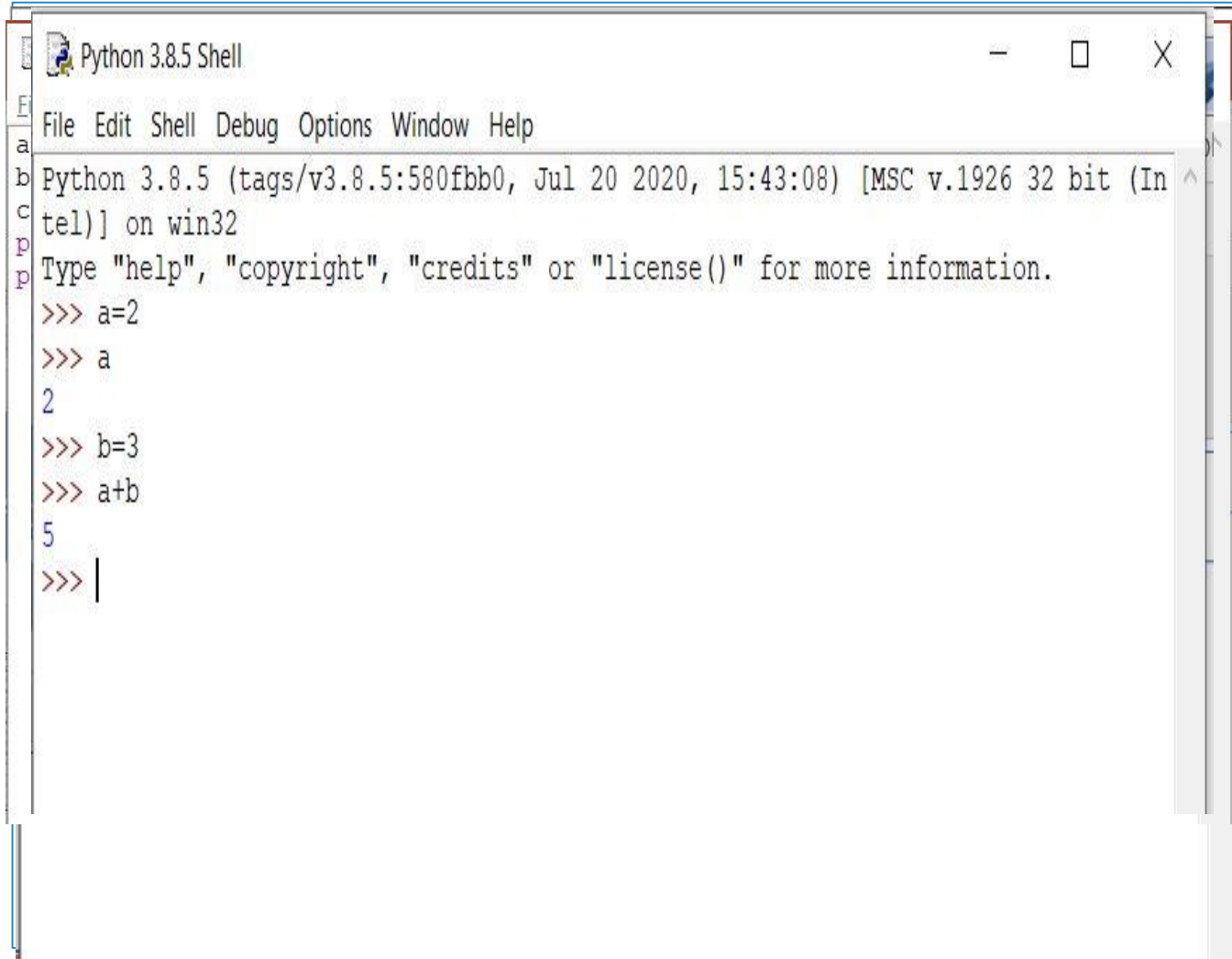
3.

INSTALLING PYTHON AND RUNNING THE PROGRAM



```
Python 3.8.5 Shell
File Edit Shell Debug Options Window Help
Python 3.8.5 (tags/v3.8.5:580fbb0, Jul 20 2020, 15:43:08) [MSC v.1926 32 bi
t (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> |
Ln: 3 Col: 4
```

- ✓ Go to website <http://www.python.org>
- ✓ Click on Downloads.
Download any Python version from the given list (python 2.7,3.2, 3.6 are the recommended versions) for windows
- ✓ Install the program
- ✓ Double clicking a python script launches a command-line window (cmd) where it runs
- ✓ You can also open IDLE (Python GUI) for script writing



The screenshot shows a 'Python 3.8.5 Shell' window with a menu bar (File, Edit, Shell, Debug, Options, Window, Help). The text inside the shell reads: 'Python 3.8.5 (tags/v3.8.5:580fbb0, Jul 20 2020, 15:43:08) [MSC v.1926 32 bit (Intel)] on win32', followed by instructions to type 'help', 'copyright', 'credits', or 'license()' for more information. The interactive session shows the following commands and outputs: '>>> a=2', '>>> a', '2', '>>> b=3', '>>> a+b', '5', and '>>> |'.

```
Python 3.8.5 Shell
File Edit Shell Debug Options Window Help
Python 3.8.5 (tags/v3.8.5:580fbb0, Jul 20 2020, 15:43:08) [MSC v.1926 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> a=2
>>> a
2
>>> b=3
>>> a+b
5
>>> |
```



- ✓ Firstly open the IDLE, go to file and create new file and then write the program and then save the program
- ✓ Then either go on run in the menu bar or else press F5 to run the program. Then it will run the program in the shell
- ✓ If we don't save the program then a dialog box opens asking to save the file and then on clicking **OK** it saves the program and then runs the program
- ✓ If you want to do small tricks then use python shell to execute

4.

DEBUGGING THE PROGRAM



DEBUGGING THE PROGRAM

- ✓ In programming various errors occurred they are called as **bugs**
- ✓ The process of tracking the bugs are called as **Debugging the program**
- ✓ There are 4 types of errors:-
 1. Syntax Errors
 2. Runtime Errors
 3. Semantic Errors
 4. Experimental Debugging

SYNTAX ERRORS

- ✓ Syntax means the structure of the program. If the syntax is not correct then it result into syntax errors
- ✓ E.g:- (2+2) this wont give an error but (2+2 then it will give an error because the parentheses is not closed hence it will give syntax errors.

RUNTIME ERRORS

- ✓ Runtime errors are known as runtime errors because the error doesn't appear until after the program has started running runtime errors are also known exceptions which indicates something bad has happened
- ✓ Runtime errors occurs very rarely

SEMANTIC ERRORS

- ✓ In case of semantic errors your program will execute properly without showing any error message but it will not give the correct result you can also say hat there is a logical error
- ✓ Tracking this error is tricky, since it requires to look for the output and find out how the code is working

EXPERIMENTAL DEBUGGING

- ✓ Debugging is a challenging work. You have to check and infer the processes and events that led to the results you see. Debugging is also known as experimental science
- ✓ If hypothesis is correct then your program will work or else you have to come up with new hypothesis



5.

BRACKETS, CURLY BRACKETS AND PARANTHESES

BRACKETS

- ✓ Brackets are used for defining the list in python.
- ✓ The list of elements are put inside the brackets.
- ✓ For example:-

list1 = [a,b,c,d,e]

list2 = [1,2,3,4,5,6]

CURLY BRACKETS

- ✓ Curly Brackets{} are used to define the dictionaries in the python.
- ✓ The elements of the dictionaries are enclosed in the curly bracket
- ✓ For example:-

Mydict= {'name':'v2v class', 'city':'Kalyan'}

PARENTHESES

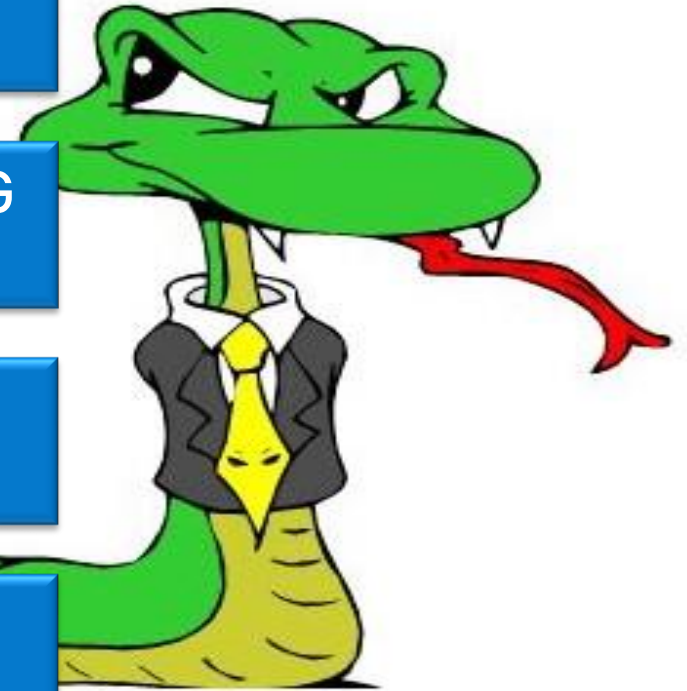
- ✓ Tuples are enclosed in the parentheses()
- ✓ The elements of the tuple are written inside the parentheses
- ✓ For example:
 $a=(1,2,3,4,5)$
- ✓ The parentheses are also used for the order of operation or evaluation of operation.

For example: $(a+b)*(c+d)$

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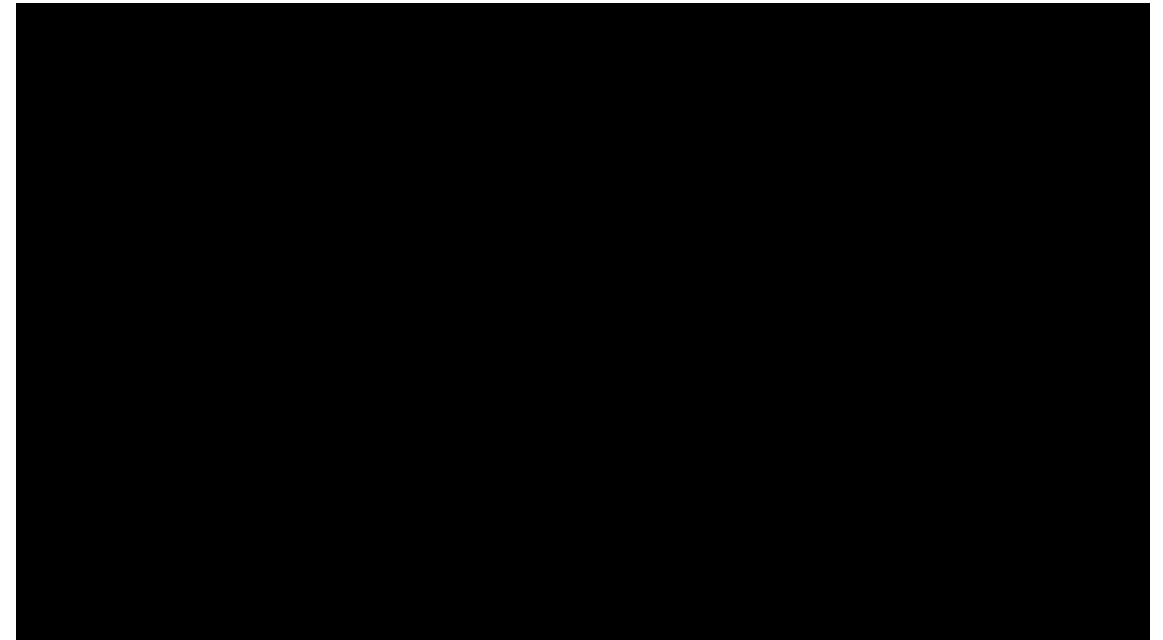
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Variables , Keywords and Type conversion

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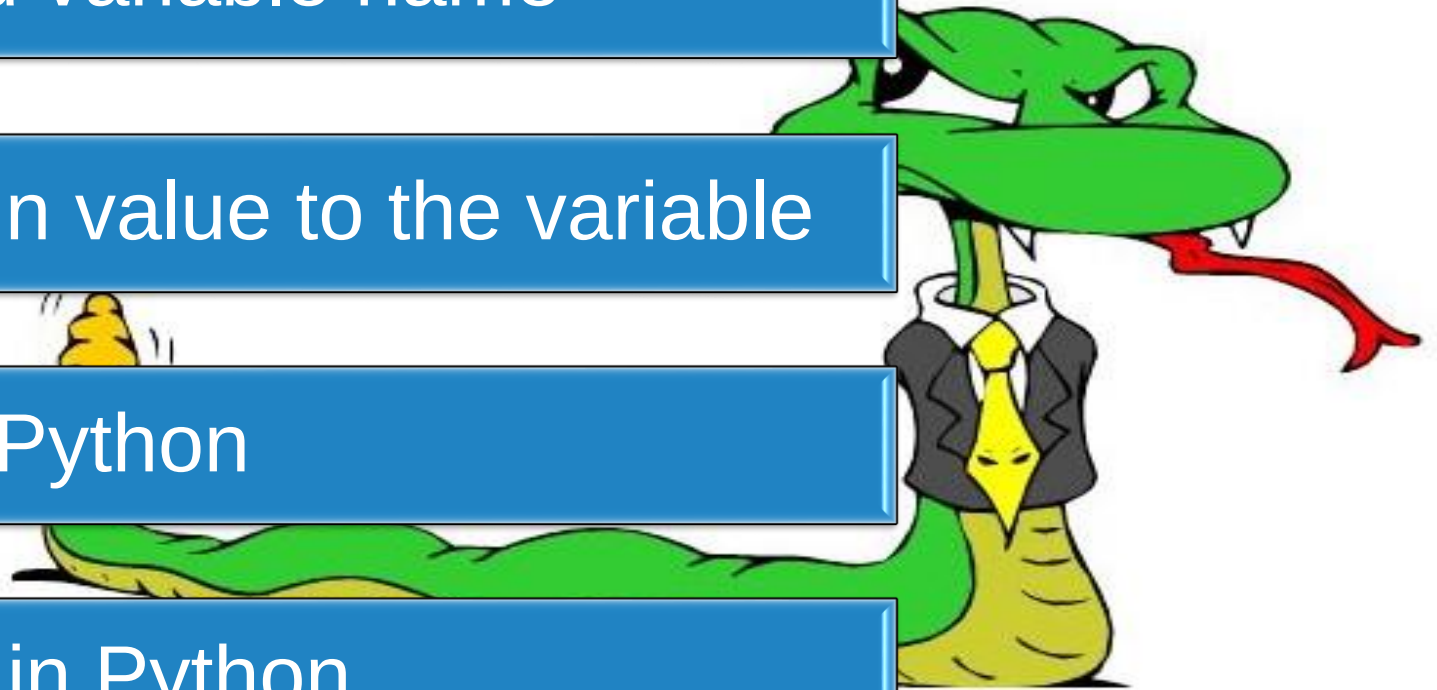
Value and Data Type

Variables and variable name

How to assign value to the variable

Key words in Python

Type conversion in Python



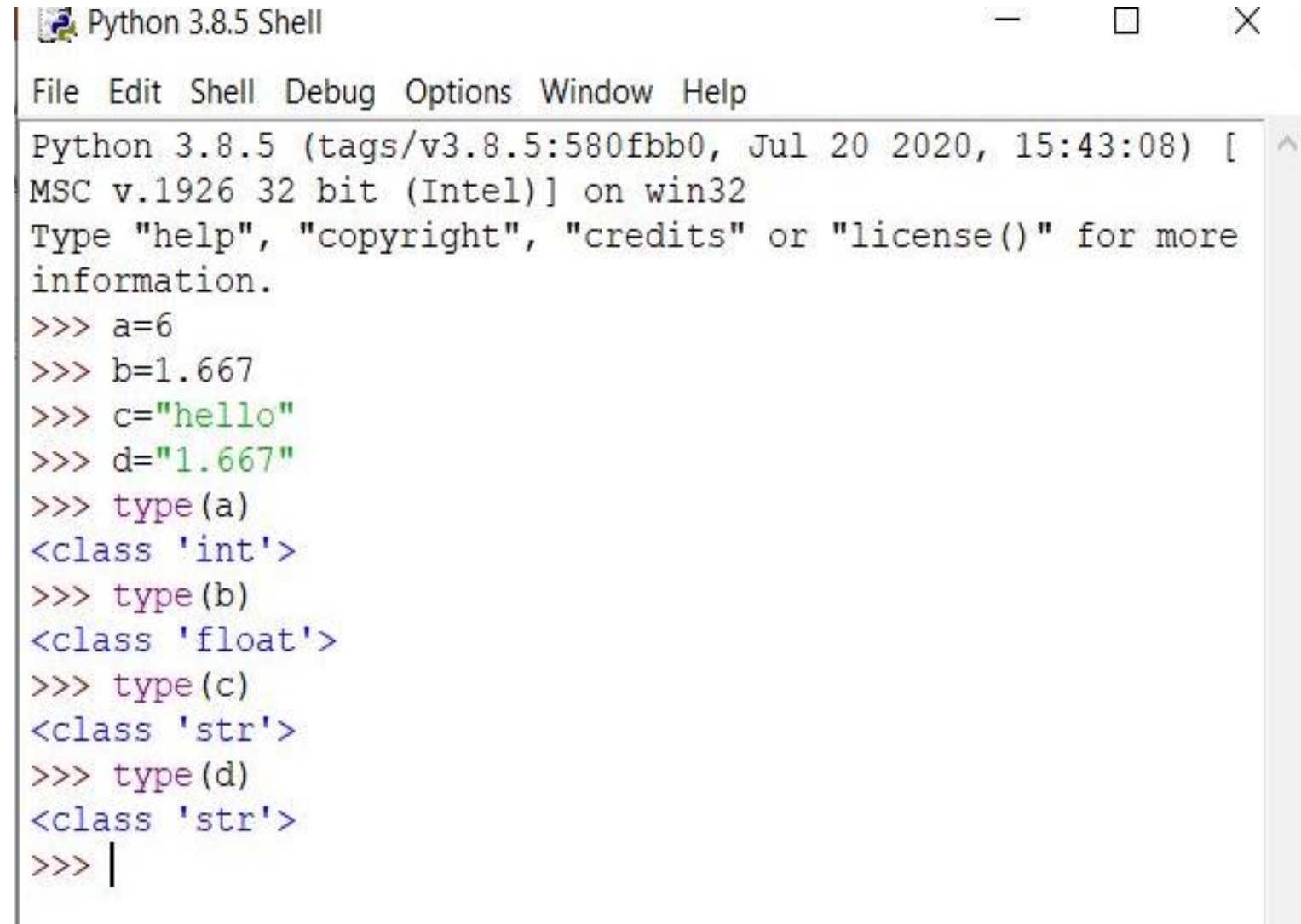
1.

VALUE AND DATA TYPE

Value and Data Types

- ✓ Value is nothing but the numbers or letters. For e.g. 1,2,"hello", 1.3222
- ✓ These value belongs to different data types
- ✓ In the above example 1 and 2 are integers
- ✓ hello is a string data type
- ✓ Any value written inside the quotation marks are considered as strings
- ✓ While 1.3222 is a float data type

- ✓ If you want to know the data type of the value then we can use **type()**
- ✓ We can see that '6' is an integer or int
- ✓ 1.667 is a float datatype
- ✓ Hello is a string datatype
- ✓ Since value of d is in quotation marks hence in output the datatype is string.



```
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Type "help", "copyright", "credits" or "license()" for more
information.
>>> a=6
>>> b=1.667
>>> c="hello"
>>> d="1.667"
>>> type(a)
<class 'int'>
>>> type(b)
<class 'float'>
>>> type(c)
<class 'str'>
>>> type(d)
<class 'str'>
>>> |
```



2.

VARIABLE AND VARIABLE NAME



VARIABLE

- ✓ Variables are reserved memory location for storing the value, when we create variable it reserves some spaces
- ✓ Depending upon the data type interpreter reserves some memory and decides what value can be stored in the allocated memory. We can store integers, floats and strings in the variables
- ✓ In python variable declaration is not required. Python determines the types of variable i.e. integers, floats, etc.

VARIABLE NAMES

- ✓ There are few rules for assigning the names to variables
 1. Variable name should begin with the letter(a-z, A-Z) or underscore(_)
 2. Other characters in variable can be letters, numbers or underscore
 3. Variable name can be of any length
 4. **Keywords** cannot be used as variable names



3.

HOW TO ASSIGN VALUES TO VARIABLES



ASSIGNING VALUE TO VARIABLES

- ✓ To assign a value to variable we use equal sign .
- ✓ Operand is on the left side of the equal sign(=) and operator is the value which is stored in the variable
- ✓ For example:- `a = 6` (in this `a` is the operand and `6` is the operator)
- ✓ Multiple Assignments:- You can assign same value to more than one variable
- ✓ You can assign different values to different variables
- ✓ For example:-
 - The variable name can be reused by assigning the new value to them.
 - You can swap the values in python in a single line

4.

KEYWORDS IN PYTHON



KEYWORDS IN PYTHON

- ✓ Keywords are reserved words and cannot be used as constant or variable or any other identifier names
- ✓ Python keywords are always in lower case letters except for 3 words
- ✓ Different keywords of python are as follows:-

True	class	finally	is	return
False	continue	for	lambda	try
None	def	from	nonlocal	while
and	del	global	not	with
as	elif	if	or	yield
Assert	else	import	pass	
break	except	In	raise	



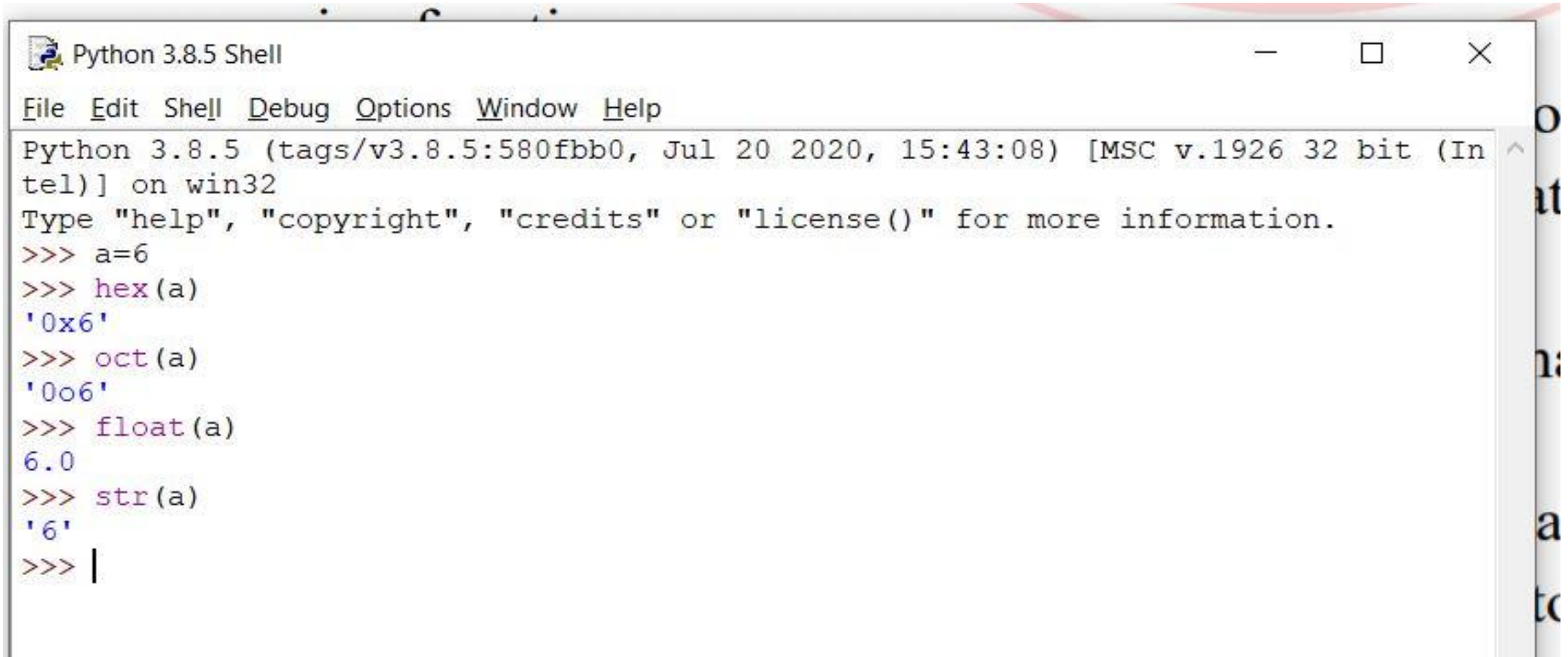
5. TYPE CONVERSION IN PYTHON



➤ Type conversion converts one data type to another data type.
.python defines type conversion features

- ☐ `int(a,base)` :- It converts any data type to integer.
- ☐ `float()` :- It converts any data type to floating point number
- ☐ `ord()` :- It converts a character to string
- ☐ `hex()` :- It converts integer to hexadecimal string
- ☐ `oct()`:- It converts integer to octal string
- ☐ `tuple()`:- It converts any list of items into tuples
- ☐ `set()` :- It return the type after converting to set
- ☐ `list()` :- It converts any data type to a list type
- ☐ `dict()`:- It converts a tuple of order(key,value) into a dictionary
- ☐ `str()` :- It converts integer into string
- ☐ `complex(real,img)` :- It converts real numbers to complex(real,img) number

Example:- The code converts integers into float, hexadecimal , octal and string

A screenshot of a Python 3.8.5 Shell window. The window has a title bar with the text 'Python 3.8.5 Shell' and standard window controls (minimize, maximize, close). Below the title bar is a menu bar with 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The main area of the window displays the following text:

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Type "help", "copyright", "credits" or "license()" for more information.
>>> a=6
>>> hex(a)
'0x6'
>>> oct(a)
'0o6'
>>> float(a)
6.0
>>> str(a)
'6'
>>> |
```

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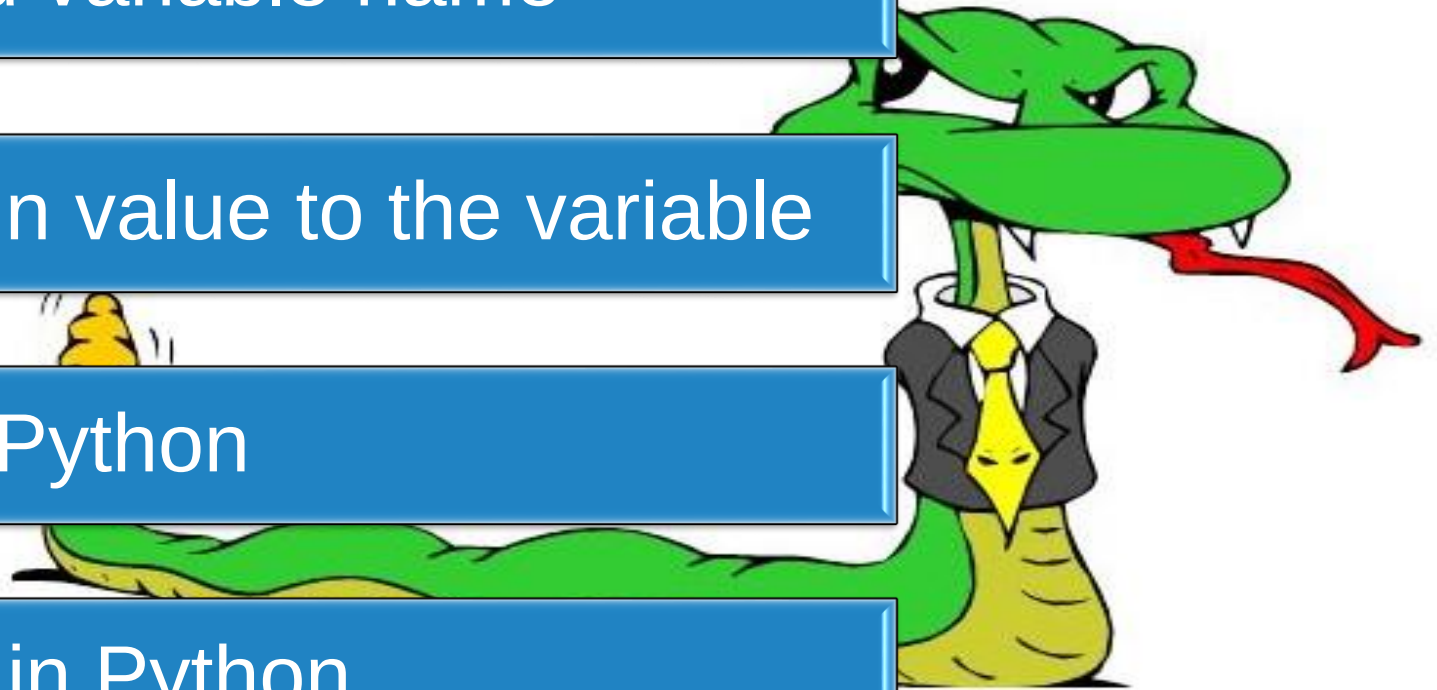
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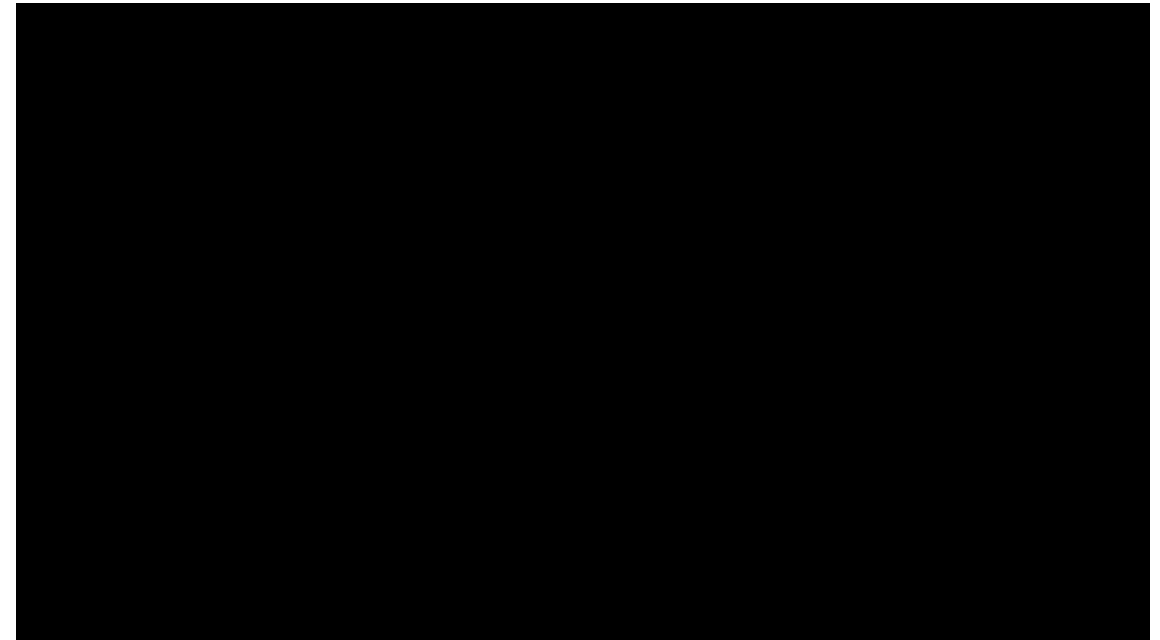
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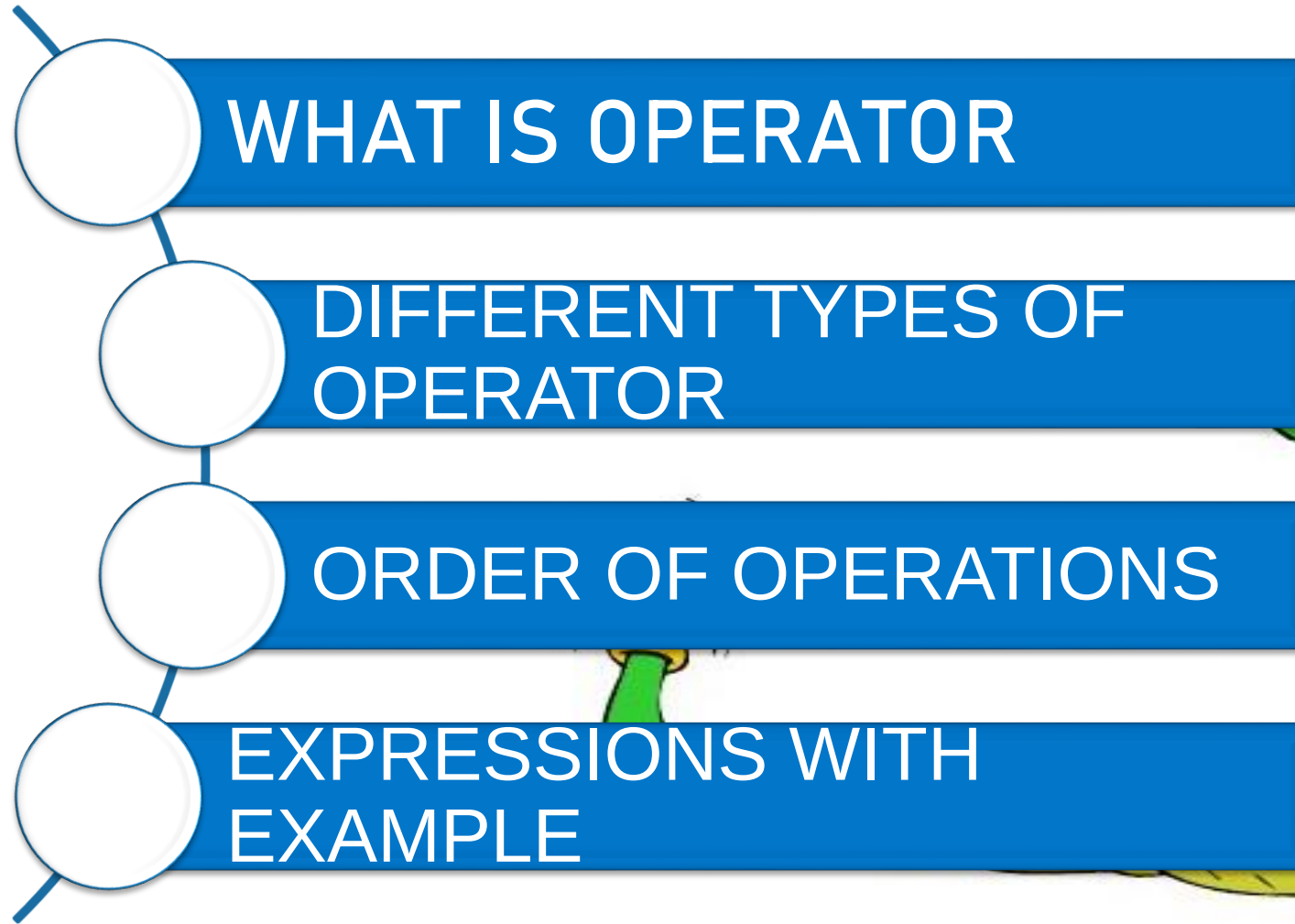
Operators in Python

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1.

WHAT ARE THE OPERATORS IN PYTHON



OPERATORS IN PYTHON

- ✓ Operators are special symbols that represents different operations
- ✓ There are different types of operators in Python:-
 1. Arithmetic Operators
 2. Comparison(Relational) Operator
 3. Assignment Operator
 4. Logical Operator
 5. Bitwise Operator
 6. Membership Operator
 7. Identity Operator



2.

DIFFERENT TYPES OF OPERATORS



1. ARITHMETIC OPERATOR

Operator	Description
+	It adds value on either side of the operator
-	It subtracts right hand operand from left hand operand
*	It multiplies values on either side of the operator
/	It divides the left hand operand by right hand operand and returns quotient
%	It divides the left hand operand by right hand operand and returns remainder
**	It performs exponential(powers) calculation on operators
//	Floor Division – The division of operands where the result is the quotient in which the digits after the decimal points are removed But if one of the operands is negative , the result is floored , i.e., rounded away from zero

2. COMPARISON OPERATOR

Operator	Description
==	If the value of the operands are equal, then the condition becomes True
!=	If value of the two operands are not equal, then the condition becomes True
<>	If value of the two operands are not equal, then the condition becomes True
>	If value of left hand operand is greater than right hand operator, then the condition becomes True
<	If value of right hand operand is greater than the left hand operand, then the condition becomes True
>=	If value of left hand operand is greater than or equal right hand operator, then the condition becomes True
<=	If value of right hand operand is greater than or equal to the left hand operand, then the condition becomes True

3 ASSIGNMENT OPERATOR

Operator	Description
=	It is used to assign the value of right side operands to left side operand
+=	It is used to add right operand to the left operand and result is assigned to the left operand.
-=	The right operand gets subtracted from the left operand and the result is assigned to the left operand
*=	The right hand operand multiplied with left operand and result is assigned to left operand
/=	Left operand is divided by right operand and result is assigned to left operand
%=	It performs modulus on two operands and the result stored to left operand
**=	It performs power calculation on operators and value is assigned to the left operand
//=	It performs floor division on operators and assigns value to the left operand

4 BITWISE OPERATOR

Operator	Description
&	It is Binary AND operator. This Operator is used to copy a bit to the result if it exists in both operands
	It is Binary OR operator .it copies the bit if it is set in one operand but not both
^	It is Binary XOR operator .it copies the bit if it is set in one operand but not both.
~	It is Binary Ones Complement operators. It is unary and has the effect of 'Flipping'
<<	It is Binary Left Shift operator .it shifts the left operands value to left by the number of bits given by the right operand.
>>	It is Binary Right shift operator. It shifts the left operands value to right by the number of bits given by the operand

5. LOGICAL OPERATOR

Operator	Description
AND	Logical AND– If both the operands are true then the condition becomes True
OR	Logical OR:- If any of the two operands are non-zero then the condition becomes True
NOT	Logical NOT :- Used to reverse the logical state of its operand.

6. MEMBERSHIP OPERATOR

Operator	Description
in	Evaluate to true if it finds a variable in the specified sequence and false otherwise.
not in	Evaluates to true if it does not finds a variable in the specified sequence and false otherwise.

7. IDENTIFY OPERATOR

Operator	Description
is	Evaluate to true if the variables on either side of the operator point to the same object and false otherwise
not is	Evaluates too false if the variables on either side of the operators point to the same object and true otherwise.

3.

ORDER OF OPERATIONS

- The order of operation means the operator precedence to evaluate an expression. The operator precedence and which are having lowest while evaluating the expression.
- The operator precedence is shown in table it is descending order

▶ Parentheses have the highest precedence and Logical OR has lowest precedence

Operator	Description
()	Parentheses
**	Exponent
+X, -X, ~X	Unary plus, Unary minus, Bitwise NOT
*, /, //, %	Multiplication, Division, Floor Division, Modulus
+, -	Addition, Subtraction
<<, >>	Bitwise Shift Operators
&	Bitwise AND
^	Bitwise XOR
	Bitwise OR
==, !=, >, >=, <, <=, is, not is, in, not in	Comparisons, Identity, Membership operators
Not	Logical NOT
And	Logical AND
Or	Logical OR

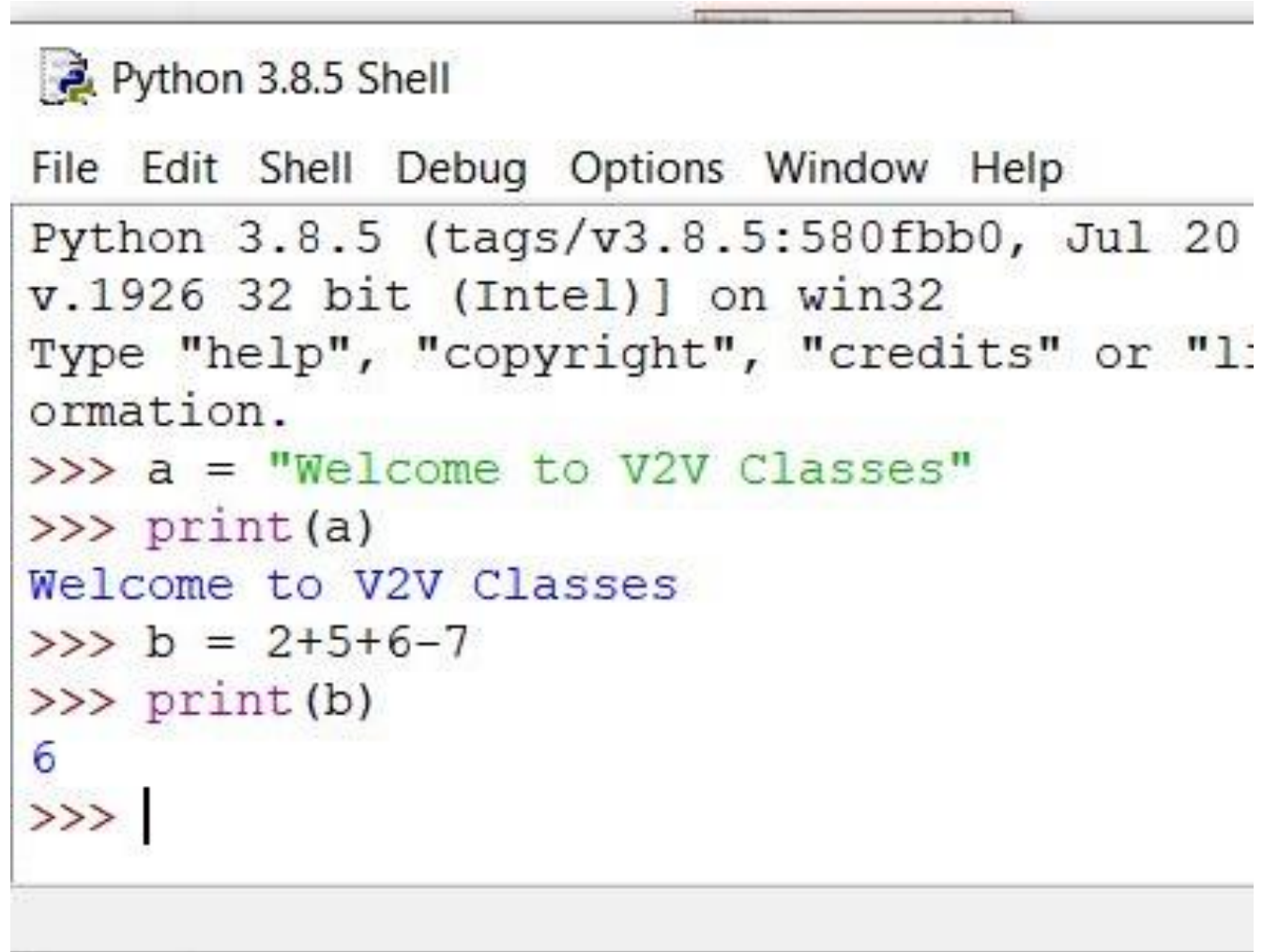
4.

EXPRESSIONS WITH EXAMPLE

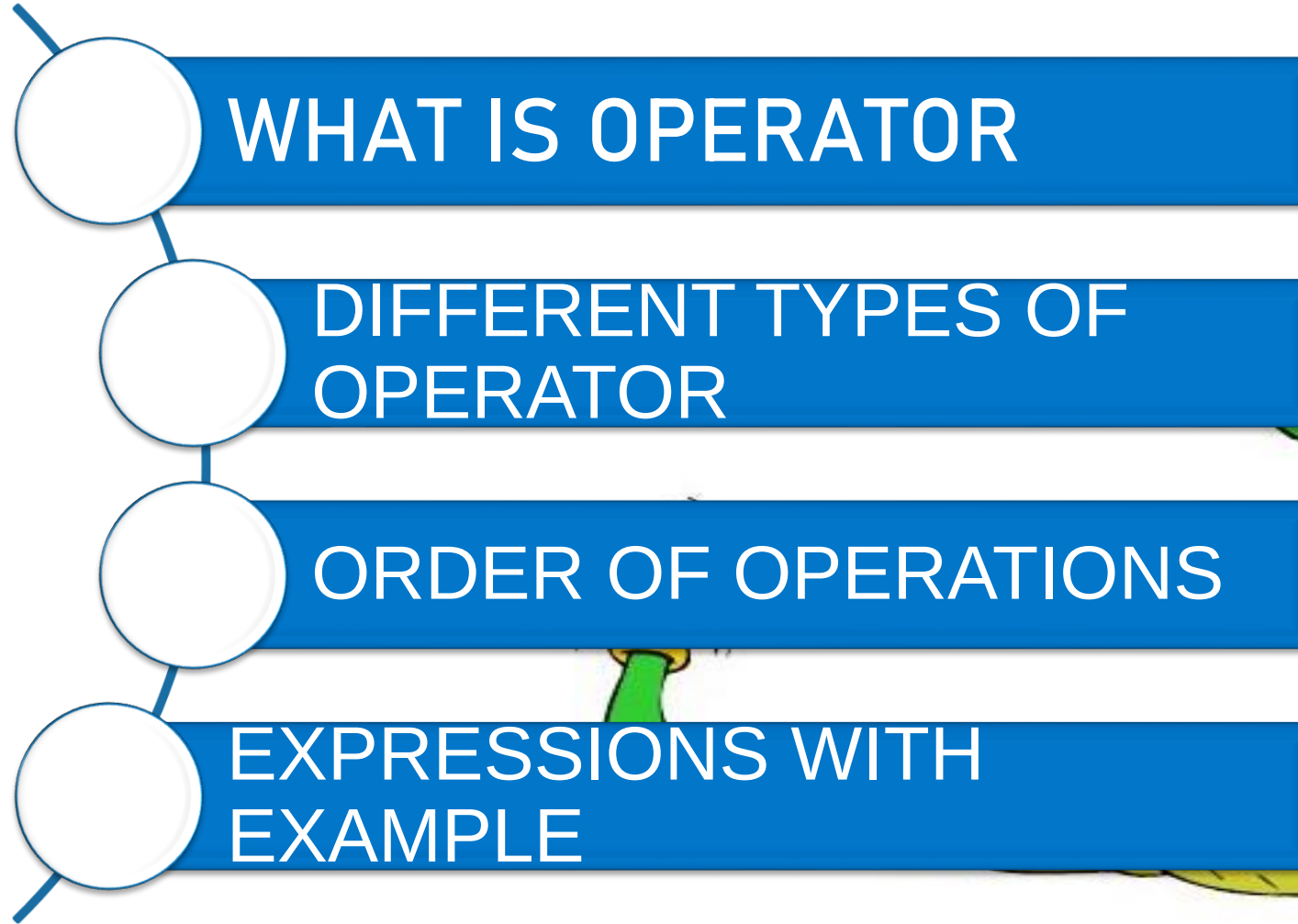


What is expression?

- ✓ An expressions is nothing but a combination of values, operators, and variables
- ✓ Evaluation of expression gives a value.
- ✓ Expressions always come at the right hand side of the assignement statements.
- ✓ Value itself is an expression and also a variable is an expression
- ✓ In the first example, string includes quotation marks and the print statement print the value inside the quotation marks
- ✓ In the second example the value that is printed is the expression value.



```
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File Edit Shell Debug Options Window Help
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v.1926 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "l:
ormation.
>>> a = "Welcome to V2V Classes"
>>> print(a)
Welcome to V2V Classes
>>> b = 2+5+6-7
>>> print(b)
6
>>> |
```



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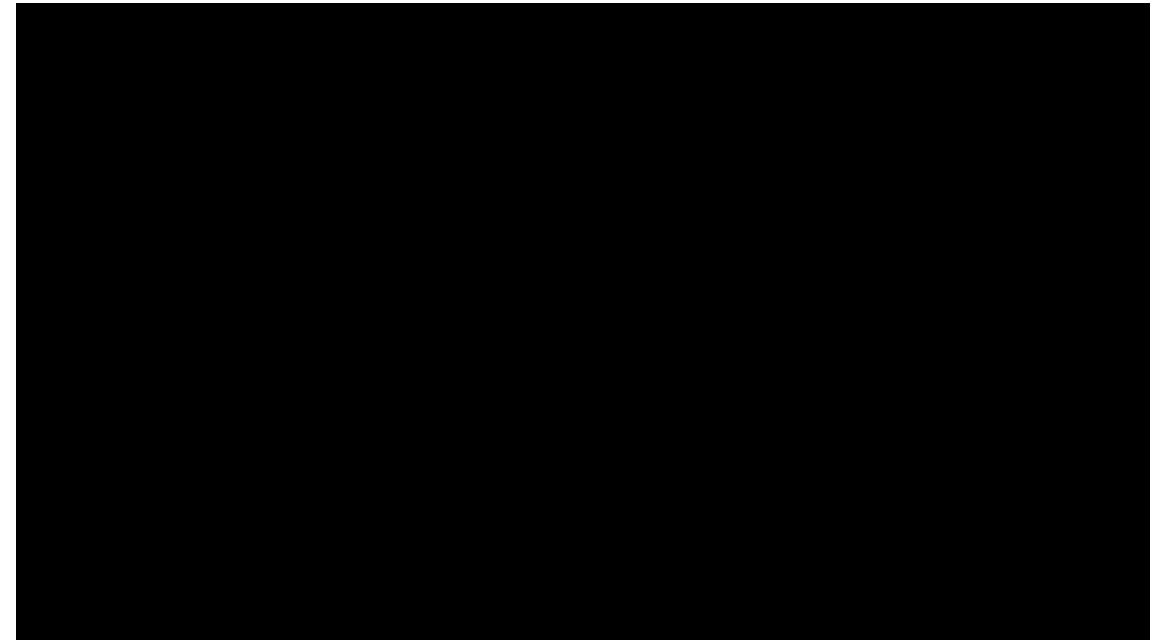
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WHAT IS IDLE IN PYTHON

INTERACTIVE MODE

SCRIPT MODE



1.

WHAT IS IDLE



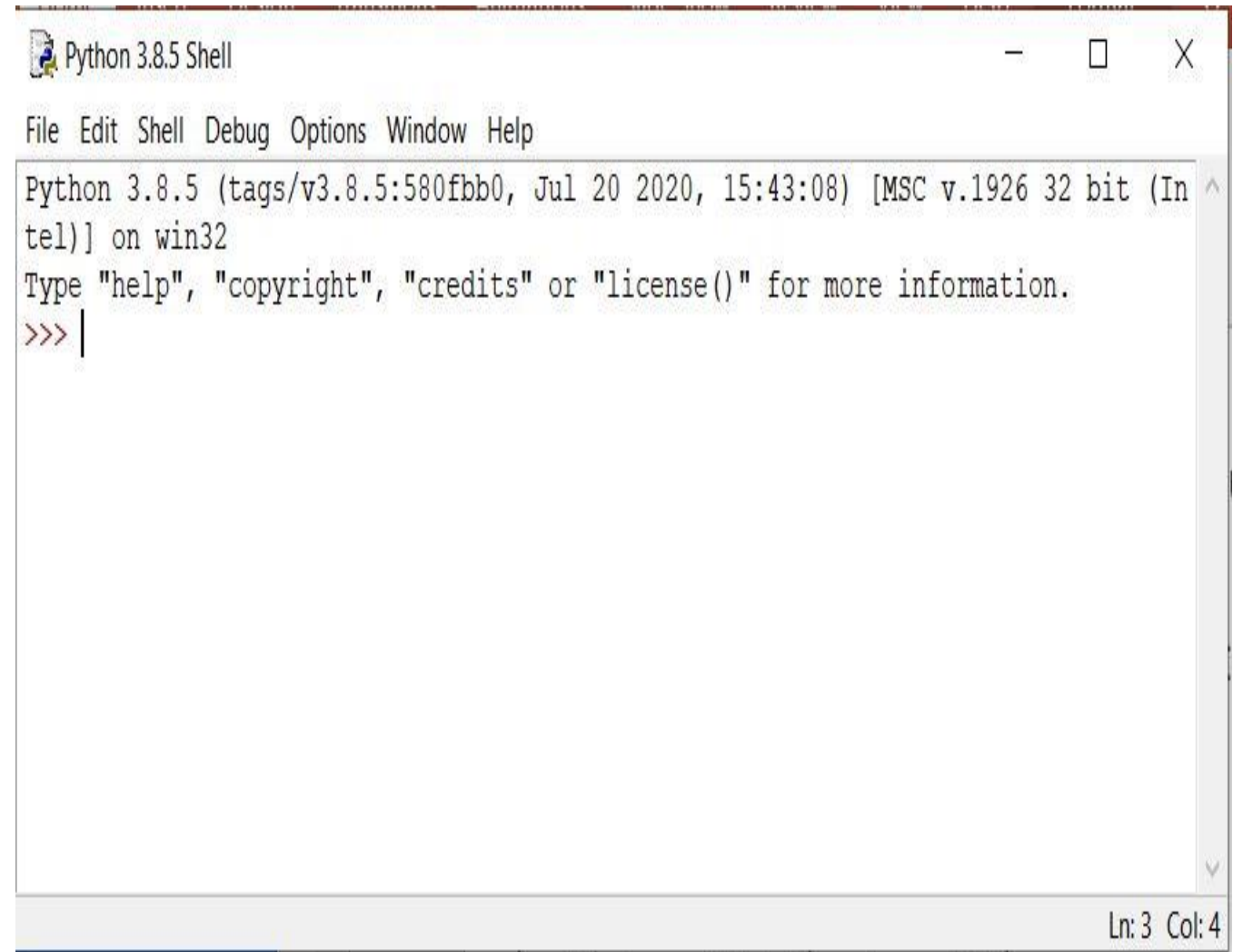
- ❑ Python comes with a Graphical User Interface(GUI) Integrated Development Environment called IDLE
- ❑ The IDLE helps the user to write the program. It enables the user to save and edit their work.
- ❑ There are Two modes of working in IDLE:-
- ❑ The TWO modes of working are:-
 1. Interactive Mode
 2. Script Mode

2.

INTERACTIVE MODE



- ✓ In interactive mode, user will tell the python what to do and python will execute the code immediately
- ✓ To open the python in interactive mode:-
 - a) Click on start menu
 - b) Click on Python version
 - c) Click on IDLE (python GUI)
- ✓ This will open the python in interactive mode
- ✓ The open window is known as Python shell also known as command prompt



```
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>>> |

Ln: 3 Col: 4
```

➤ Understanding Jargons or special words/ expression

- ✓ print() function is known as statement
- ✓ Python is case sensitive, so all commands of python are in lowercase. For e.g. if we write print function as PRINT or Print then python will generate an error message

➤ Errors Generation

- ✓ If computer does not understand any command which is typed incorrectly will generate the error
- ✓ E.g Print('v2v classes') → In this you can see that the print command is in uppercase, it is not understood by the computer, hence it generates the error

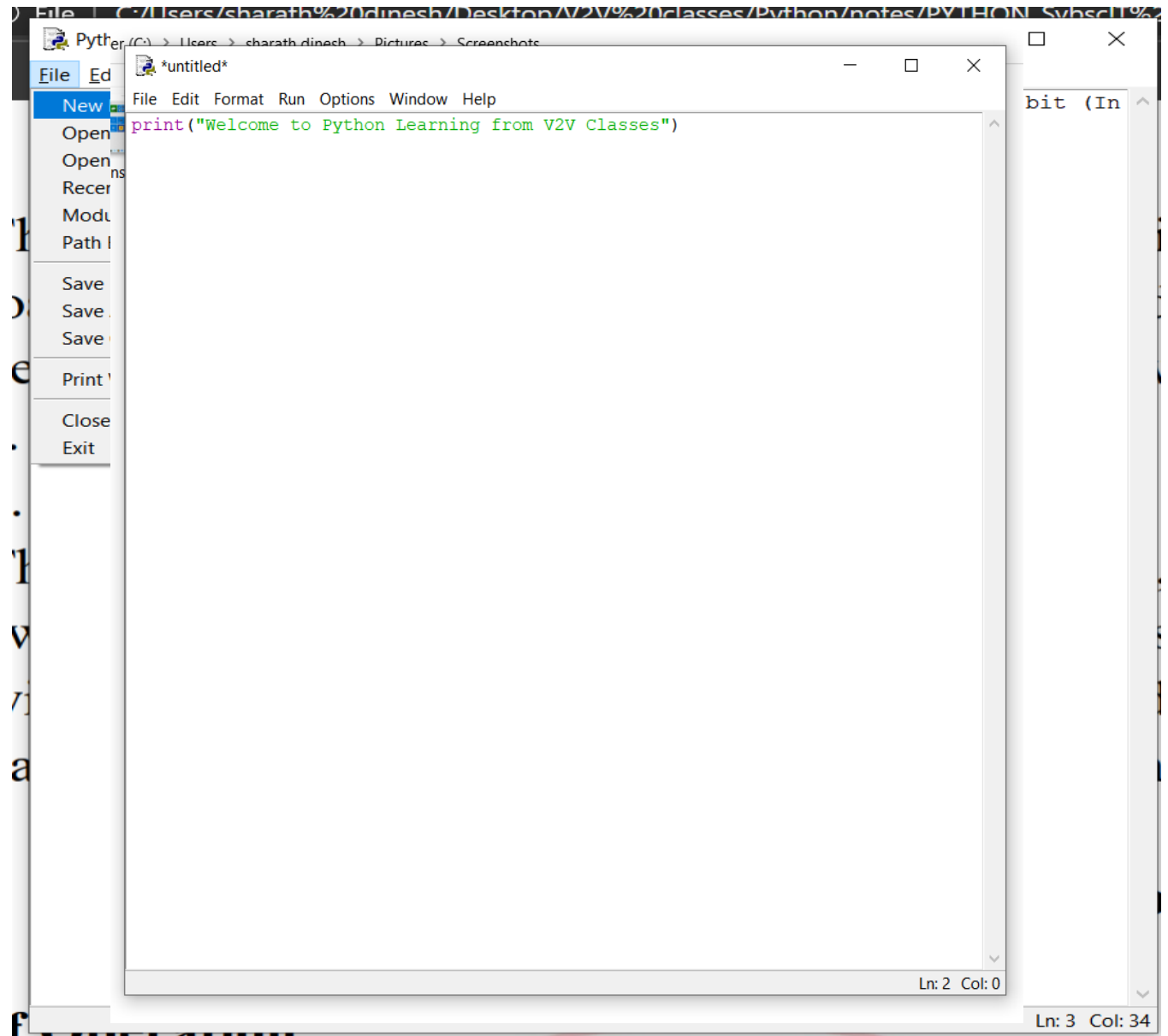
➤ Colour Coding

- ✓ Python uses the colour coding which will be helpful to understand what you have typed visually categorizing it.
- ✓ For e.g.:- print("Welcome toV2V Classes")
- ✓ In this print is in pink colour and the message written inside the parentheses is in green colour and the output is in blue colour

3. SCRIPT MODE



- ✓ Second mode of IDLE is script mode.
- ✓ In this mode we can write, edit, load and save your Python program like word processor
- ✓ You can also cut, copy and paste operations
- ✓ To open the window in script mode:
 1. Click on file menu on the interactive window
 2. Click on New File
- ✓ After this a script window will get open
- ✓ If you will write `print("welcome to python learning in v2v classes")` and press ENTER key then nothing will happen
- ✓ Because we are not in interactive mode.
- ✓ We have to save the program first and then run it



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WHAT IS IDLE IN PYTHON

INTERACTIVE MODE

SCRIPT MODE



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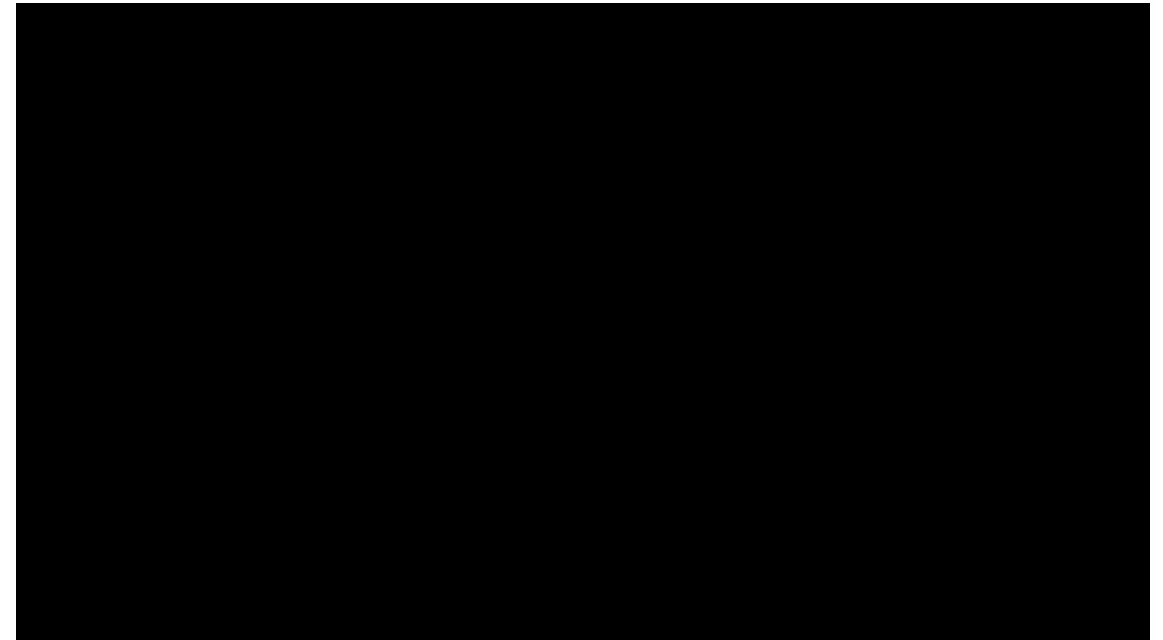
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Conditional Statements in Python

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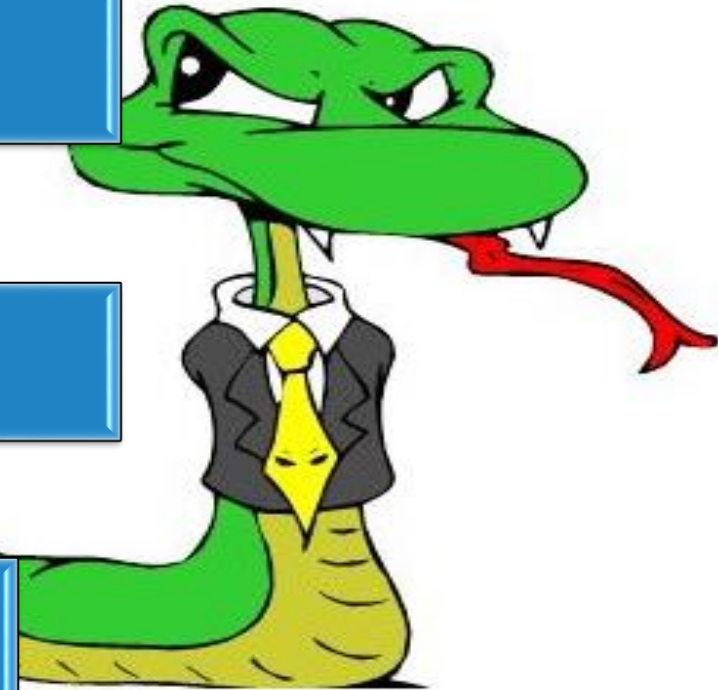
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What is conditional statement and its types

If else statement

Nested if else

Elif ladder



1.

WHAT IS CONDITIONAL STATEMENTS AND ITS TYPES IN PYTHON

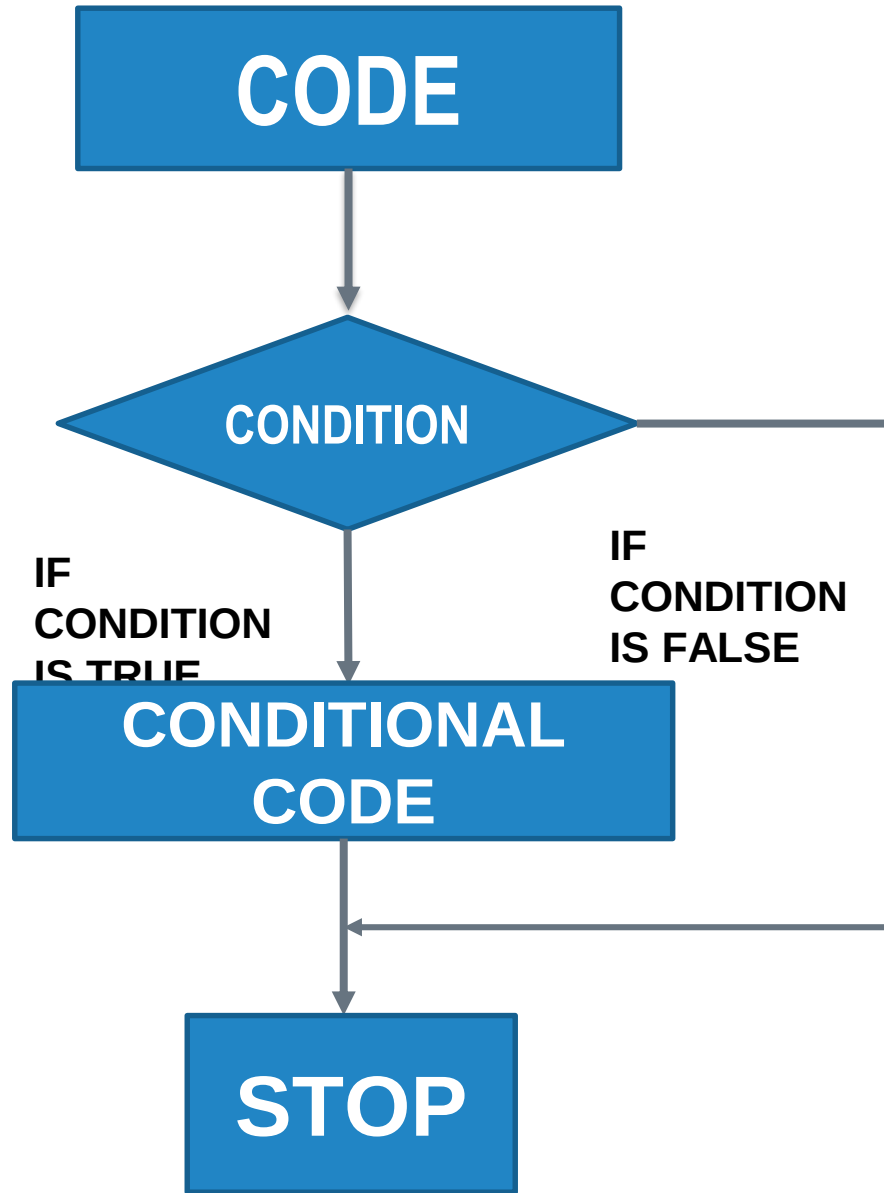
How does this
Conditional
statement works

- In 'Python' program is e
- If you put some condition on the result evaluated 'Python.
- Conditional statements
- These conditions are sp which are evaluated to :

THESE

- ☐ IF ELSE STATEMENT
- ☐ NESTED IF ELSE STATEMENT
- ☐ IF- ELSE-IF LADDER STATEMENT



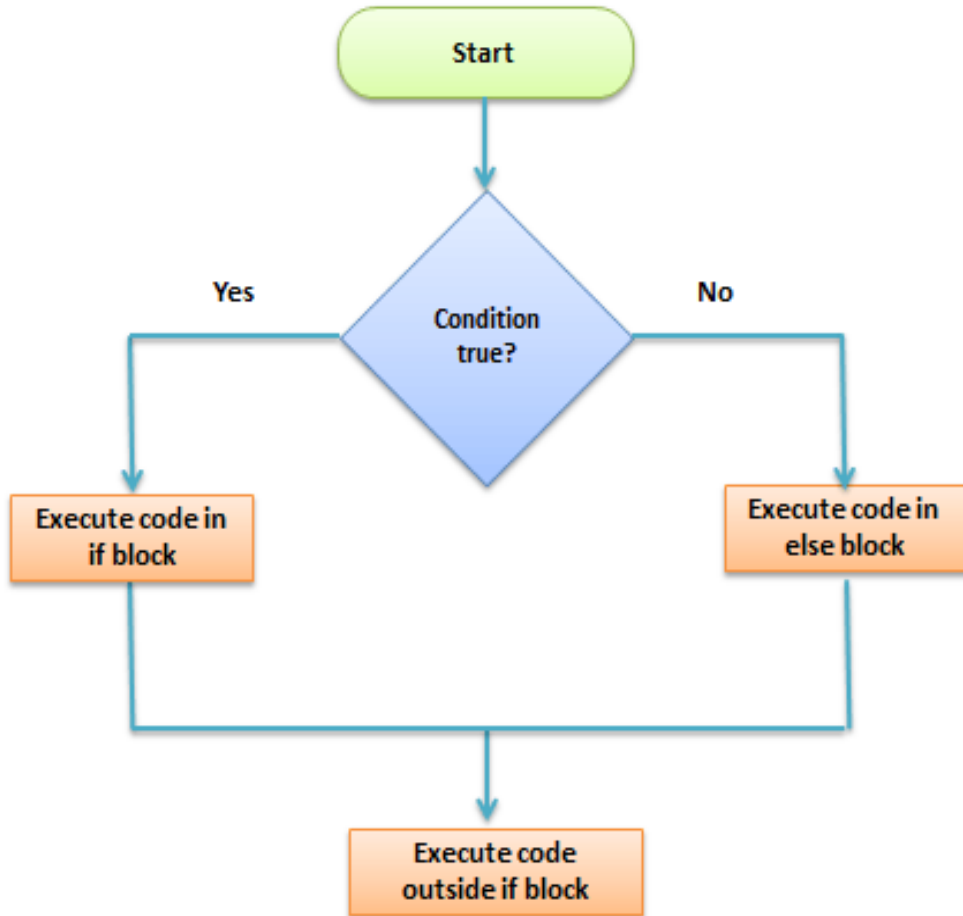


- Firstly, the main function is executed and during execution the compiler encounters the if loop
- Then it checks the condition for the Boolean value like true and false.
- Consider the value is true, then the compiler will first compile the conditional code
- Then it again execute the main function
- Now if we consider the Boolean value is False
- then it will not execute the conditional code.
- It will directly execute the main function after the if-else block

2.

IF ELSE STATEMENTS

- IF statement is the most powerful conditional statement.
- In this statement we use relational operator i.e. ==,>,<,<=,>=,!=,etc.
- If the condition in IF block becomes False then the ELSE block is executed



- For example:
we know that if an angle is 90° then the triangle is right angle
Now we will see it in IF ELSE form
if(angle==90)
 triangle is right angle
else
 triangle is not right angle
- NOTE:- If else loop is even possible without an else block

SYNTAX

If condition:
 statement body1
else:
 statement body2

3.

NESTED IF ELSE STATEMENTS

- If we require a series of decision, then we use nested IF-ELSE
- Use of one IF-ELSE statement inside an IF statement or ELSE statement is called as Nested IF ELSE statement

```
if boolean_expression_1:  
    if nested_expression_1:  
        statement 1  
    else :  
        statement 2  
else:  
    if nested_expression_1:  
        statement 3  
    else:  
        statement 4
```

NOTE: In nested if-else, we have to be careful with the indentation because multiple if-else constructs are involved in this process, so it becomes difficult to figure out individual constructs. Proper indentation makes it easy to read the program.

4.

ELIF STATEMENTS

- When a programming situation requires the choice of one case from many different cases successive if statements can be tied together forming what is sometimes called as if-else-if statements
- An if condition is tested only when all previous if conditions in if-else ladder is false
- If any of the conditional expression evaluates to true, then it will execute the corresponding code block and exits whole if-else ladder.

SYNTAX

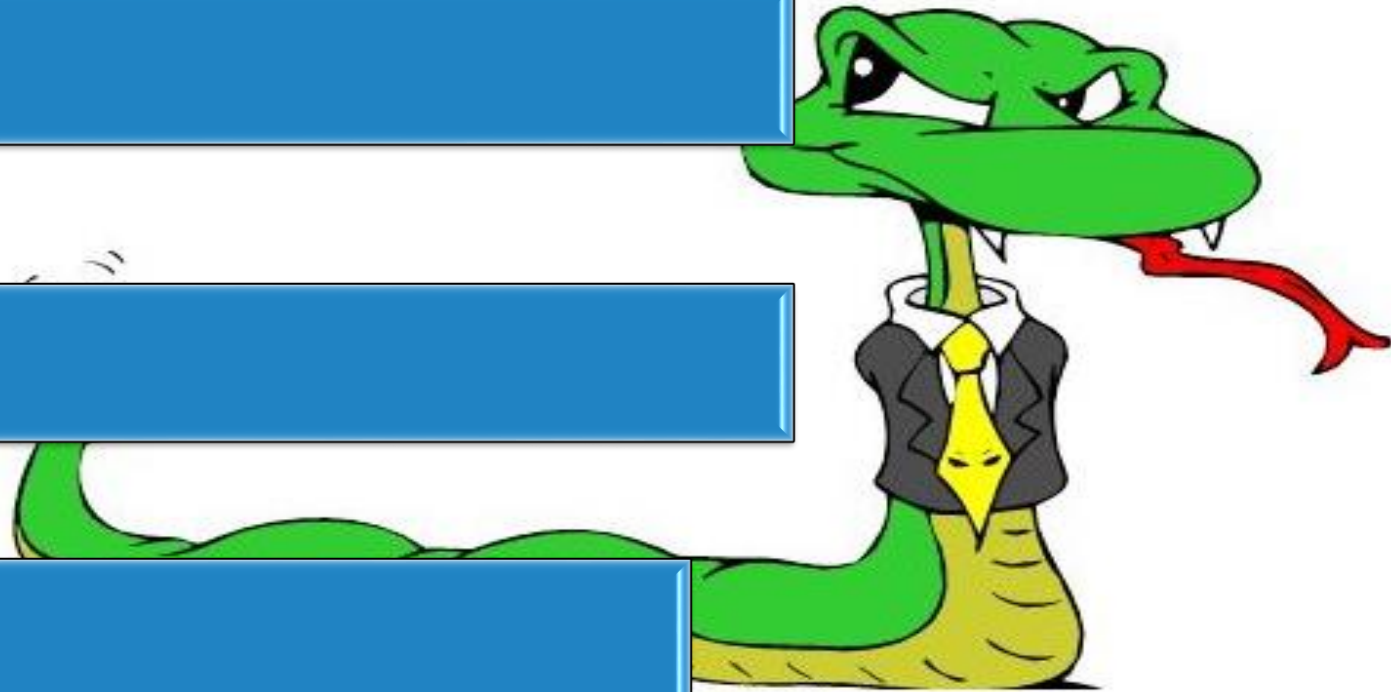
```
If condition1 :  
    statement1  
elif condition2:  
    statement2  
elif condition3:  
    statement3  
else:  
    statement4
```

- First of all condition1 is tested and if it TRUE then statement1 will be executed and control comes out of the whole if-else ladder
- If condition1 is FALSE then condition2 is tested.
- Control will keep on flowing downward, if none of the conditional statement is TRUE.
- The last else is the default block of code which will gets executed if none of the conditional statement is TRUE

Python

Capsule

- What is conditional statement and its types
- If else statement
- Nested if else
- Elif ladder



THANK YOU



Prof. Akshay Gaikwad

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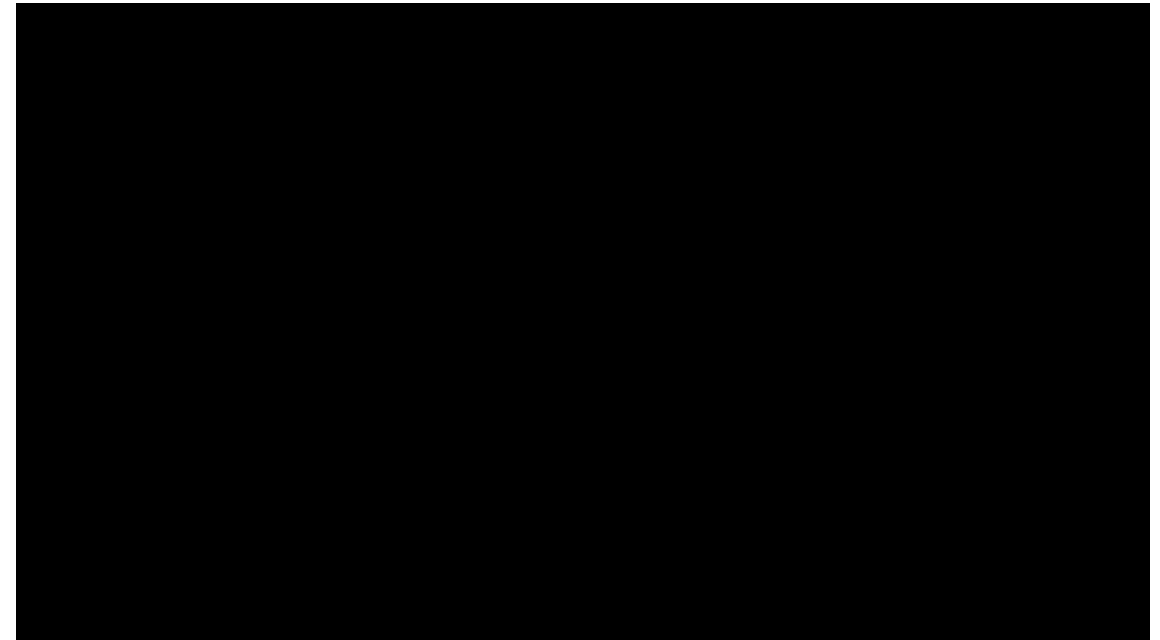
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Loops in Python

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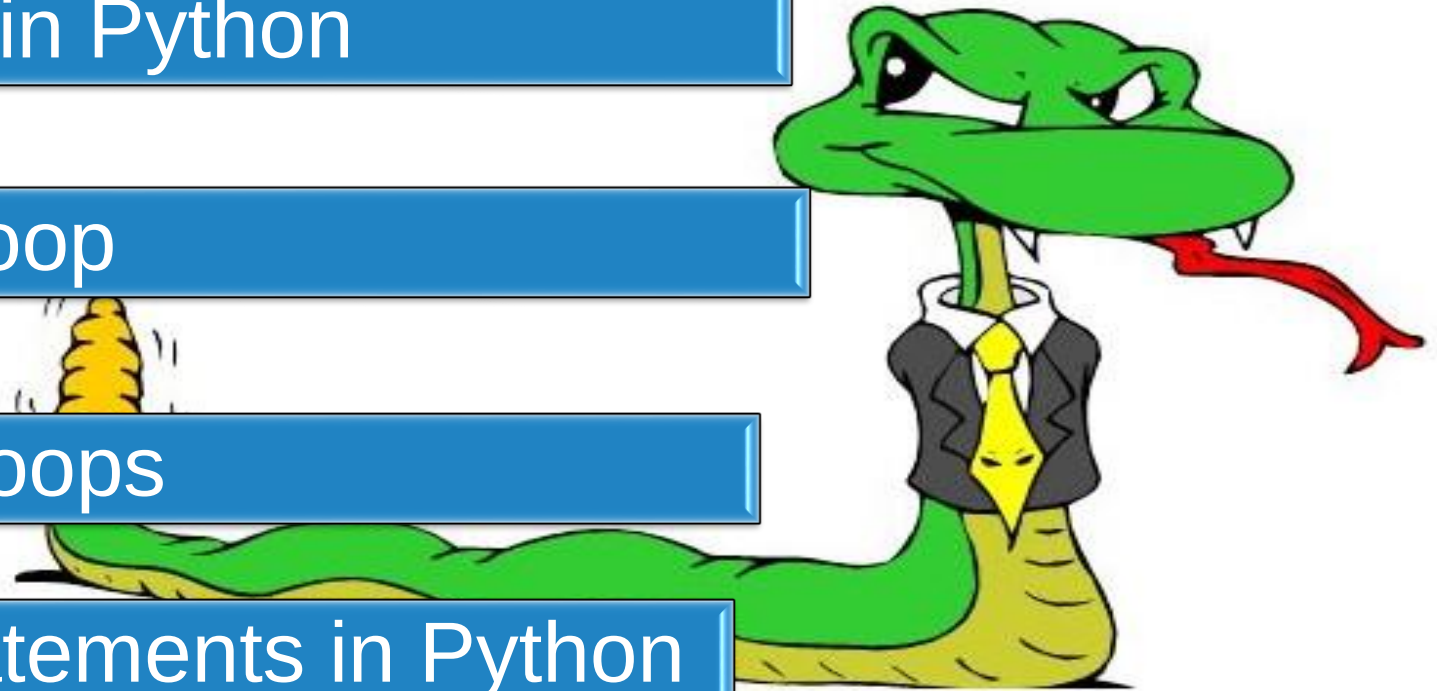
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Python

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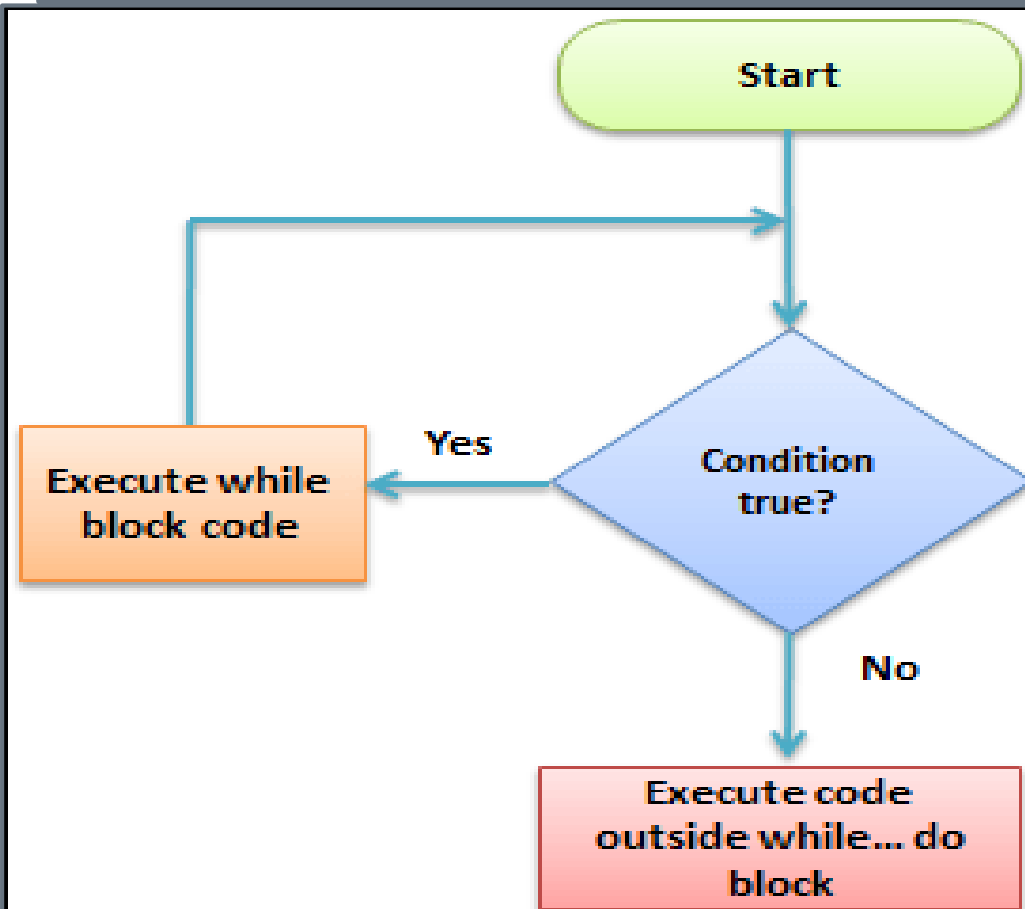
- What are loops and its types
- For Loop in Python
- While Loop
- Nested Loops
- Control Statements in Python



1.

WHAT ARE LOOPS IN PYTHON AND ITS TYPES

- A Loop executes the sequence of statements many times until the stated condition becomes false.
- A loop consists of two parts, a body of a loop and a control statement. The control statement is a combination of some conditions that direct the body of the loop to execute until the specified condition becomes false.
- The purpose of the loop is to repeat the same code a number of times



There are 3 types of Loops in Python:-

- FOR LOOP
- WHILE LOOP
- NESTED FOR LOOP



2.

FOR LOOP IN PYTHON

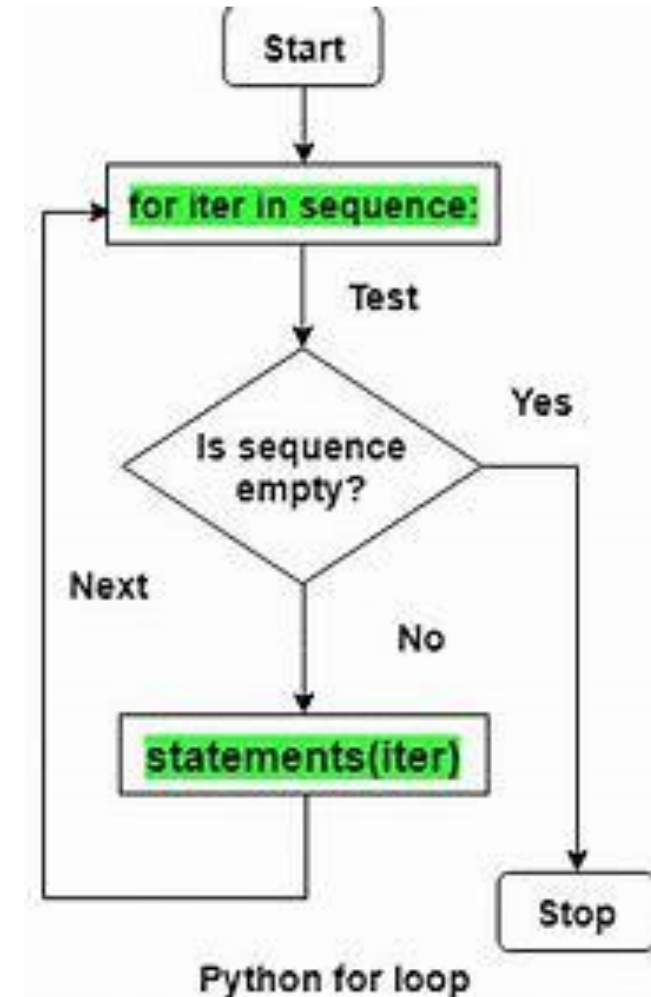
A **for** loop is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times.

SYNTAX

for iterating `_va` in sequence:

statements;

- If the Logical expression list contained by a sequence then it is evaluated first .The first element in the sequence is assigned to the iterating variable i.e. iterating `_var` After that the block of the statement is executed until the complete sequence is done or finished
- The `range ()` function is a built –in function. it iterates over a sequence of numbers . It produces an iterator of arithmetic series.
- The `range ()` function produce an iterator to progress integers begin with 0 upto `n-1`. You can typecast the `range ()` to `list ()` to get a list object of the sequence. After that you can iterate the list by using a `for` Statement.
- Another way of iterating through each item is by using index offset into the sequence itself. The following example shows iterating through each item or element using index using index offset
- Using `else` Statement with Loops:
 - Python allows you to use conditional statement along with the loop statement
 - You can use the `else` statement with the `for` loop



3.

WHILE IN PYTHON

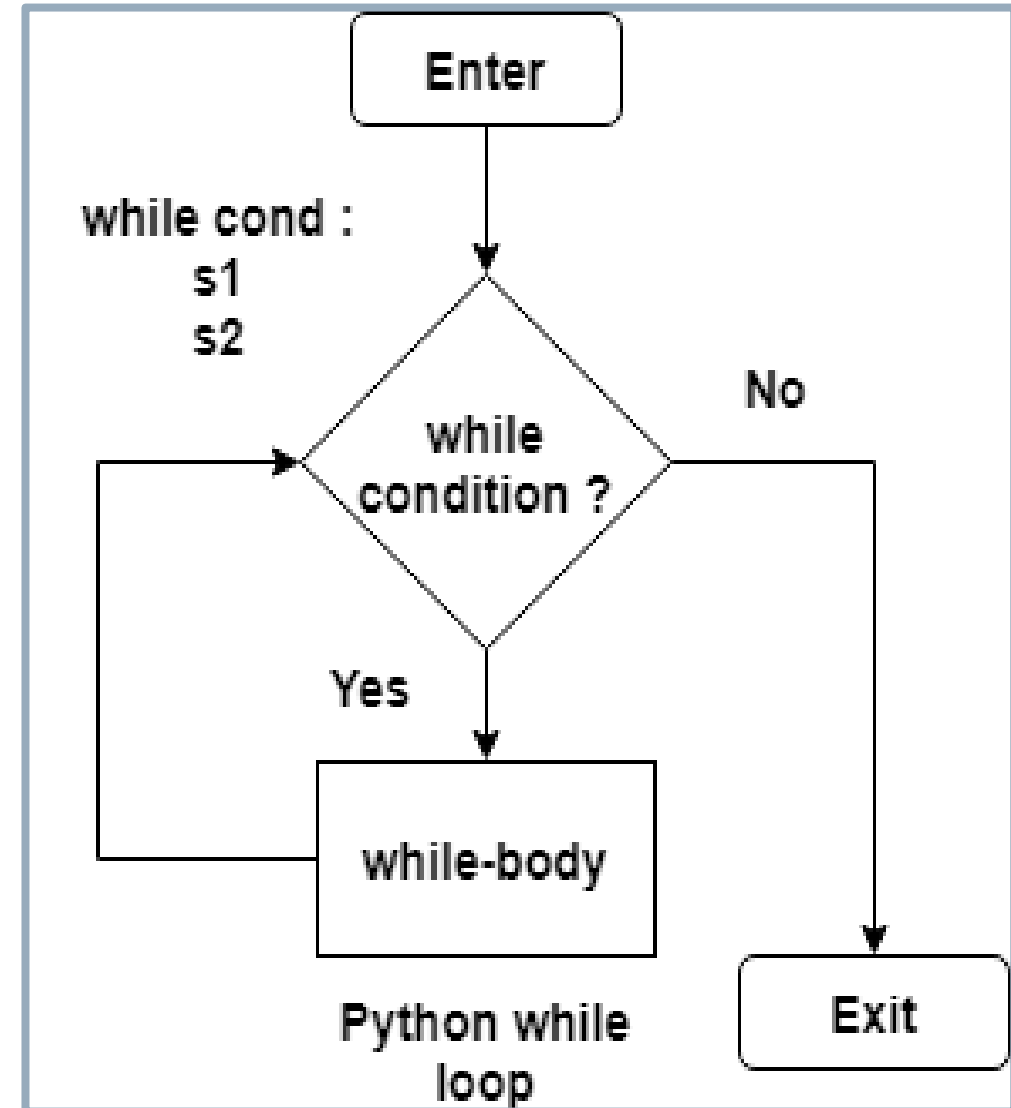
A **while** loop in Python programming repeatedly executes a target statement as long as a given condition is true.

SYNTAX

While condition:
statements

STEPS HOW THE WHILE LOOP EXECUTES

- In the syntax the statement can be single or multiple statements of block of statements with uniform indent
- The condition may be any Logical expression.
- The condition is true if any non-zero value is given. Till the condition become true the while loop iterates. If the condition returns true then the statements inside the body of while loop are executed else control comes out of the loop.
- If the condition becomes false then the program control passes to the line immediately following the loop..
- Python uses indentation for grouping the statements. All the statement are by same number of character spaces



INFINITE LOOP

- ✓ When the condition never become false then loop become the infinite loop.
- ✓ You have to take care when using the while loop, because there is possibility that the condition will never resolves to a false value.
- ✓ The infinite loop means the result in the loop never ends.
- ✓ You can use the infinite loop in client server programming. In client server programming the server needs to run continuously so that client programs can communicate with the server whenever it required.
- ✓ As you can see the output program has gone into the infinite loop. To exit from the infinite loop press CTRL +C Keys .
- ✓ Using else statement with loops
 1. Python is having the feature to use else statement with the loop statement .
 2. With the for loop if the else statement is used , the else statement is executed when the loop has done in iterating the list .
 3. With the while loop if the else statement is used , the else statement is executed if the condition becomes false.

4.

NESTED LOOPS IN PYTHON



- ✓ Python allows you to use nested loops that is one inside the another loop. You can use for loop inside the for loop .
- ✓ The for loop which is inside finishes first and then it passes the control to the outer for loop statement.

SYNTAX

Following is the syntax of a nested for loop :-

```
for iterating_var1 in sequence1:  
    for iterating_var2 in sequence2:  
        statement(s)  
    statement(s)
```

Following is the syntax of a nested while loop:-

```
while expression1:  
    while expression2:  
        statement(s)
```

In loop nesting you can also put any type of loop inside any other type of loop.

It means you can put a **for** loop inside a **while** loop or vice-versa

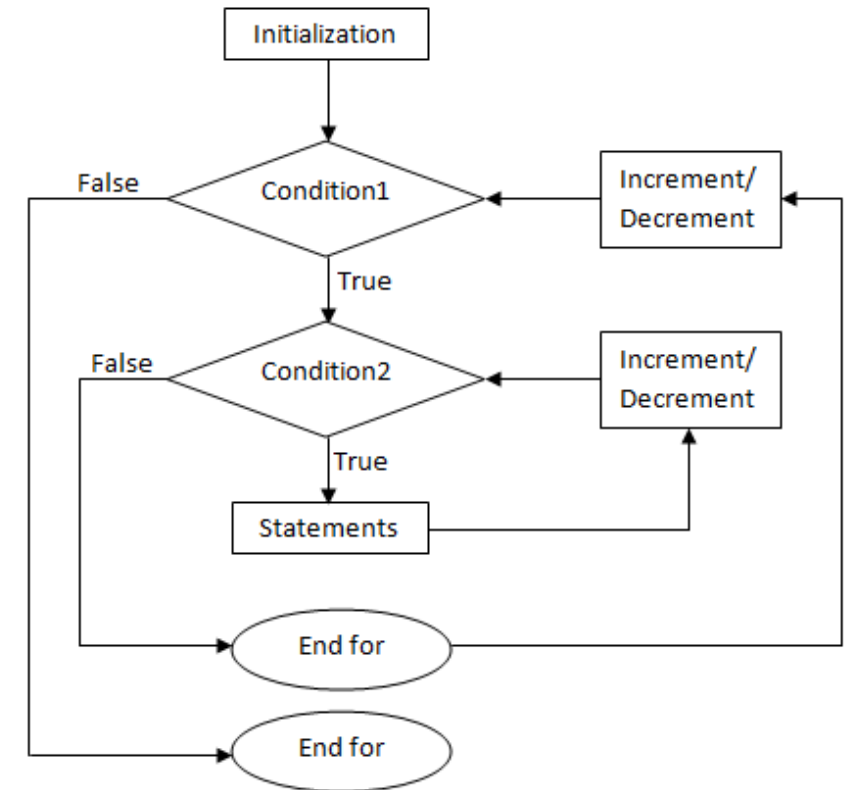


Fig: Flowchart for nested for loop

5.

CONTROL STATEMENTS IN PYTHON



WHAT IS CONTROL STATEMENTS IN PYTHON?

- ✓ Loop control statements change execution from its normal sequence.
- ✓ When execution leaves a scope, all automatic objects that were created in that scope are destroyed.
- ✓ Python supports the following control statements.:
 1. Break
 2. Continue
 3. Pass

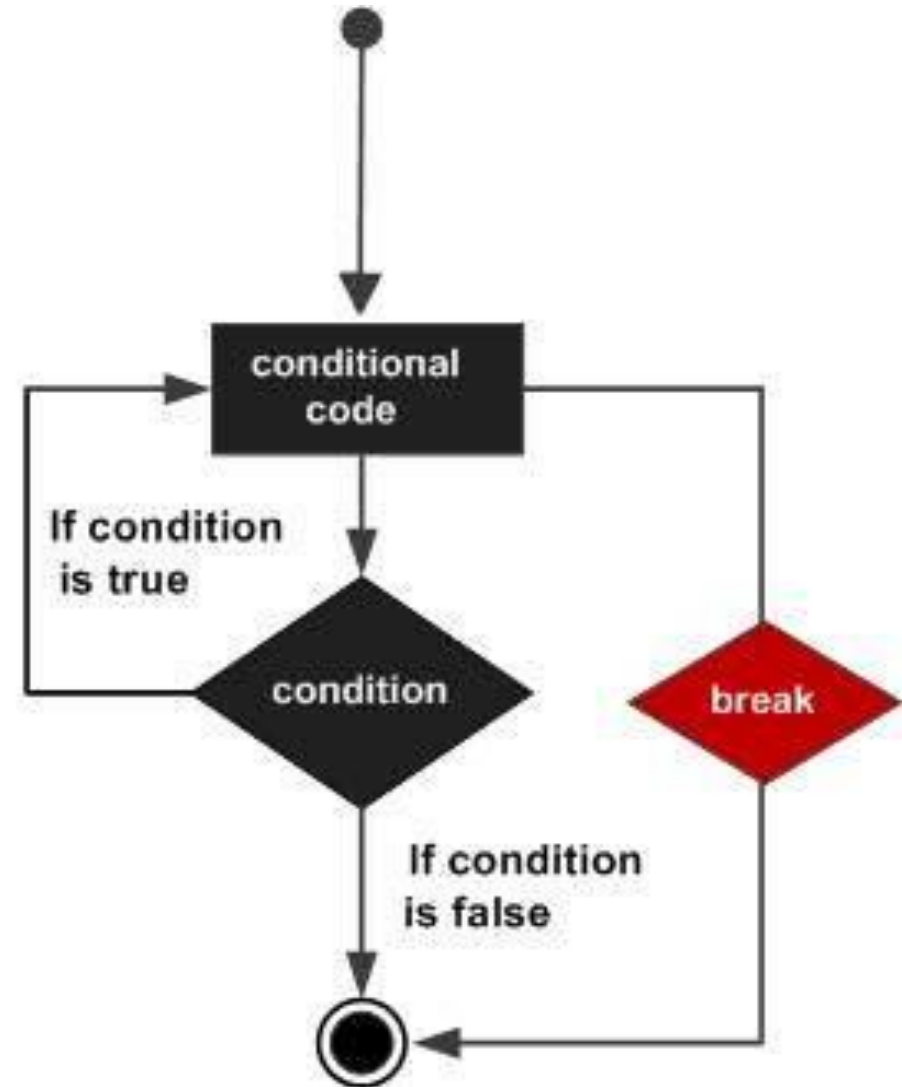


Break Statements

- ✓ Break statements is used to terminate the loop and it transfer execution to the statement immediately following the loop.
- ✓ If the loops are nested then the break would only exit from the loop which contains it. It means the break statement exit only from the single loop
- ✓ The break statement's most common use is when any external condition triggered requires a quick exit from the loop . you can use the break statement in for and while loop.
- ✓ Following is the syntax of break statement :

Syntax:-

Break

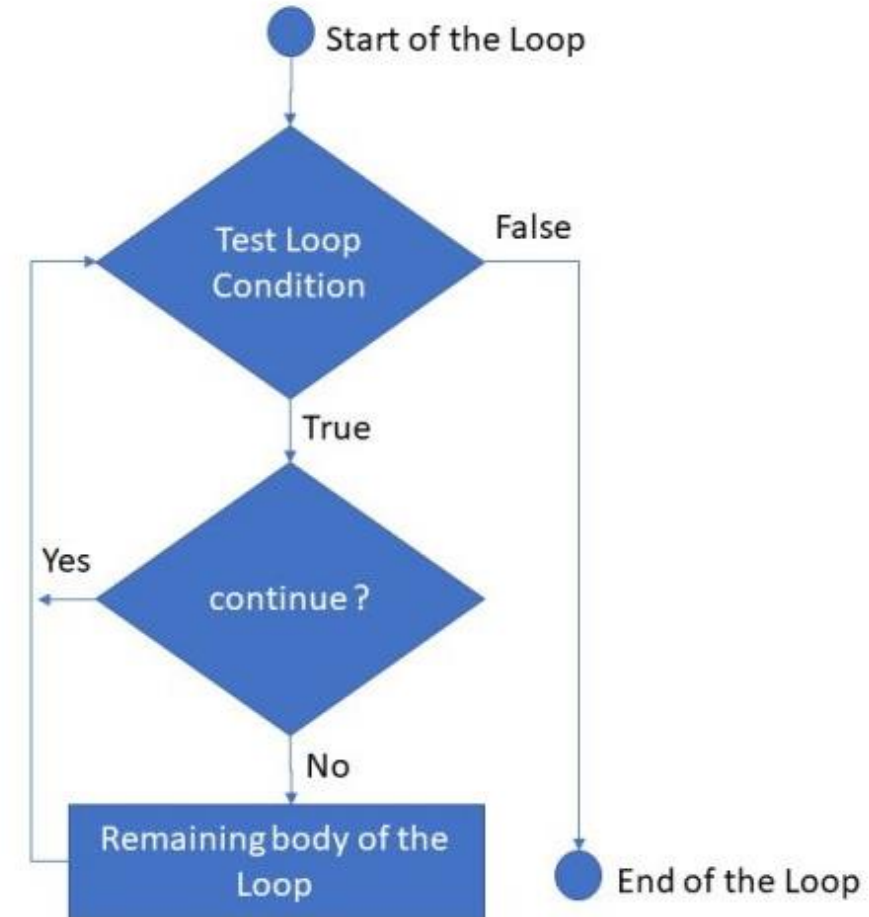


Continue Statements

- ✓ The continue statement makes the loop to skip the rest of its body and immediately reset its condition before the re-iteration.
- ✓ The continue statement returns the control to the start of the current loop.
- ✓ When the condition encountered, the loop do not execute the remaining statements and starts the next iteration
- ✓ The continue statement is used in both for while loop.
- ✓ Following is the syntax of continue statement:

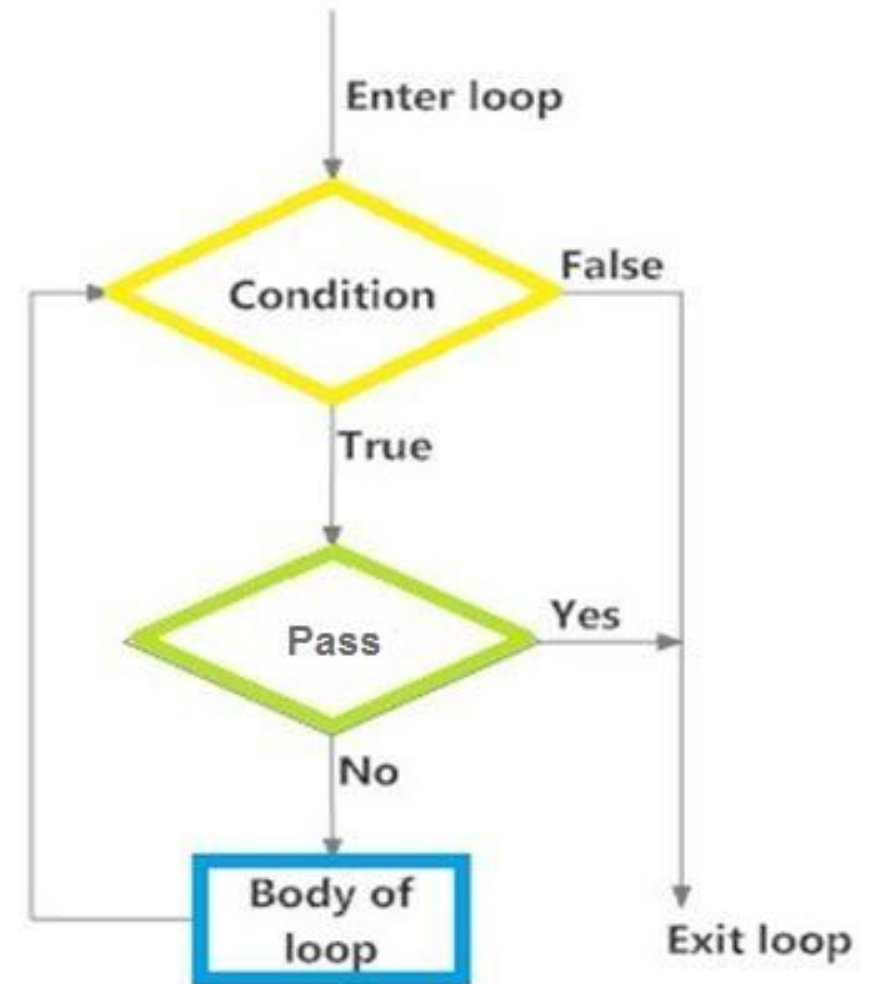
Syntax

Continue



Pass Statements

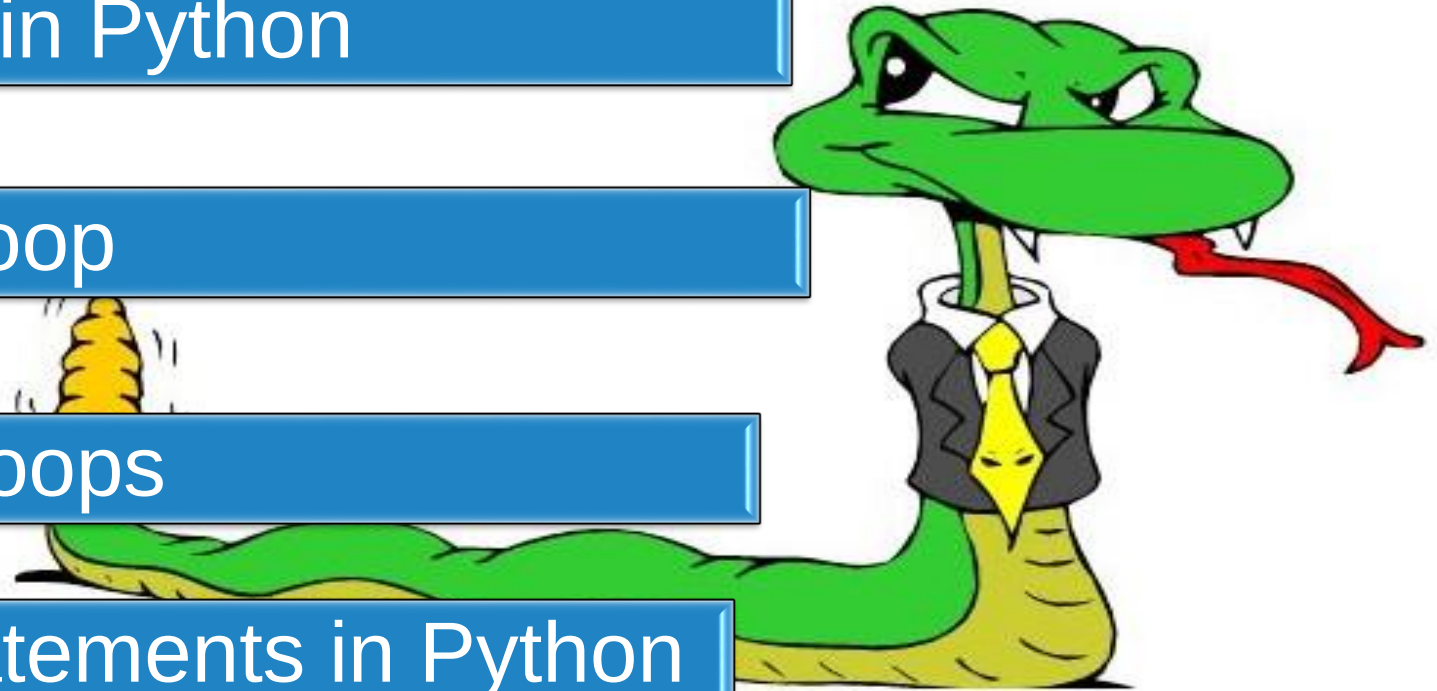
- ✓ If there is a need of syntactical statement yet you don't need any other or code to execute then you can use the pass statement.
- ✓ When you execute then pass statement nothing will happen so you can say the pass statement is an invalid or null operation.
- ✓ The pass statement is valuable in place where your code will in the end go , yet has not been composed yet. For instance stubs.



Python

Capsule

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